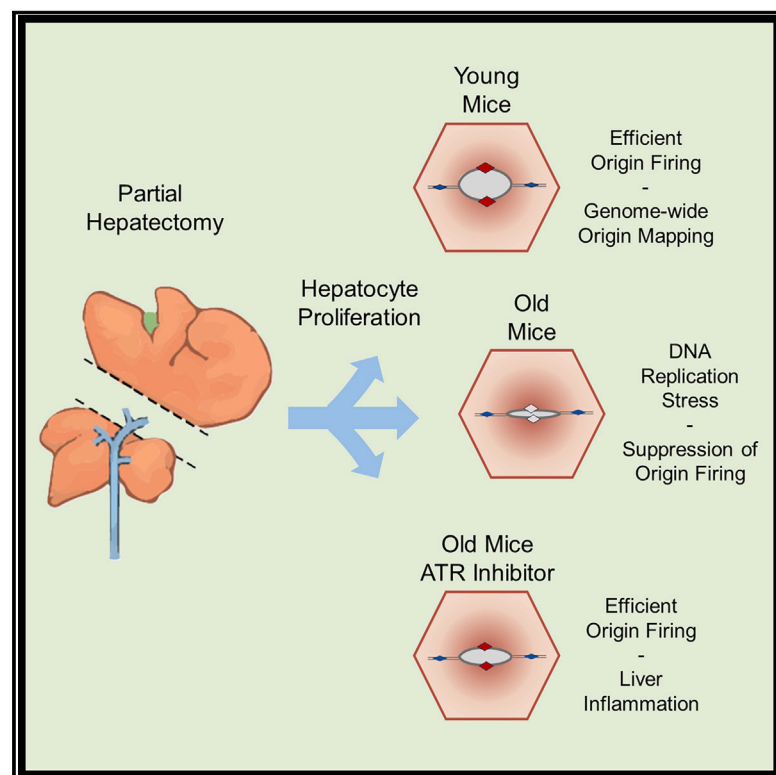


***In vivo* DNA replication dynamics unveil aging-dependent replication stress**

Graphical abstract



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In brief

In vivo DNA replication dynamics in regenerating mouse livers unveil aging-dependent decline in replication efficiency and a crucial role of the ATR checkpoint kinase in mitigating inflammation associated with the decline.

Highlights

- Partial hepatectomy enables *in vivo* mapping of DNA replication origins in mouse livers
- Most origins are located 10–50 kb upstream or downstream of expressed genes
- An aging-dependent reduction in origin firing is mediated by the ATR checkpoint
- ATR inhibitors restore origin firing in old mice but lead to liver inflammation



Article

In vivo DNA replication dynamics unveil aging-dependent replication stress

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SUMMARY

The genome duplication program is affected by multiple factors *in vivo*, including developmental cues, genotoxic stress, and aging. Here, we monitored DNA replication initiation dynamics in regenerating livers of young and old mice after partial hepatectomy to investigate the impact of aging. In young mice, the origin firing sites were well defined; the majority were located 10–50 kb upstream or downstream of expressed genes, and their position on the genome was conserved in human cells. Old mice displayed the same replication initiation sites, but origin firing was inefficient and accompanied by a replication stress response. Inhibitors of the ATR checkpoint kinase fully restored origin firing efficiency in the old mice but at the expense of an inflammatory response and without significantly enhancing the fraction of hepatocytes entering the cell cycle. These findings unveil aging-dependent replication stress and a crucial role of ATR in mitigating the stress-associated inflammation, a hallmark of aging.

INTRODUCTION

Duplication of the mammalian genome is a critical process, whose high fidelity is important to ensure normal development and healthy aging and to minimize the onset of cancer and other diseases.^{1–14} Thousands of DNA replication origins are needed to duplicate mammalian genomes, and significant effort has been invested to map their positions on the genome. These efforts, performed mostly with cells grown in tissue culture, have utilized widely different experimental approaches and cell types.^{15–18} While some consensus is emerging, given the great diversity in samples and methods used, it is not surprising that the origin firing sites identified by the various studies do not overlap perfectly.^{19–30} Notably, whether these *in vitro* findings reflect replication dynamics *in vivo* remains unknown.

Since mapping the positions of DNA replication origins is important for understanding how mammalian genomes are organized and replicated, we attempted to map origin positions *in vivo* using regenerating mouse livers as a model system. In adult mammals, the hepatocytes do not proliferate, but they can be induced to enter S phase rather synchronously after partial hepatectomy (PH).^{31–33} The liver may therefore be an ideal organ to study origin firing *in vivo*, because we can obtain both non-prolif-

erating and synchronized proliferating tissues that are needed to generate and validate origin mapping data.

As described below, we were able to obtain highly reproducible DNA replication data from regenerating mouse livers. In young mice, we observed efficient origin firing. In contrast, analysis of regenerating livers from old mice revealed that aging led to highly inefficient origin firing, associated with induction of a DNA damage response. Remarkably, inhibiting the DNA replication stress checkpoint kinase ATR allowed efficient origin firing in the regenerating livers of old mice, but it induced an inflammatory response. Our findings represent the first *in vivo* examination of DNA replication dynamics under physiological conditions, unveiling the impact of aging. They also provide insights on how fidelity of replication of the genome is ensured.

RESULTS

Firing of DNA replication origins *in vivo* after PH

Since PH leads to a relatively well-synchronized entry of hepatocytes into S phase,^{31–33} we could employ the 5-ethynyl-2'-deoxyuridine sequencing (EdU-seq)-hydroxyurea (HU) method to identify replication origins.^{27,34} At several time points after PH, mice were injected with EdU, a thymidine analog, and HU; EdU



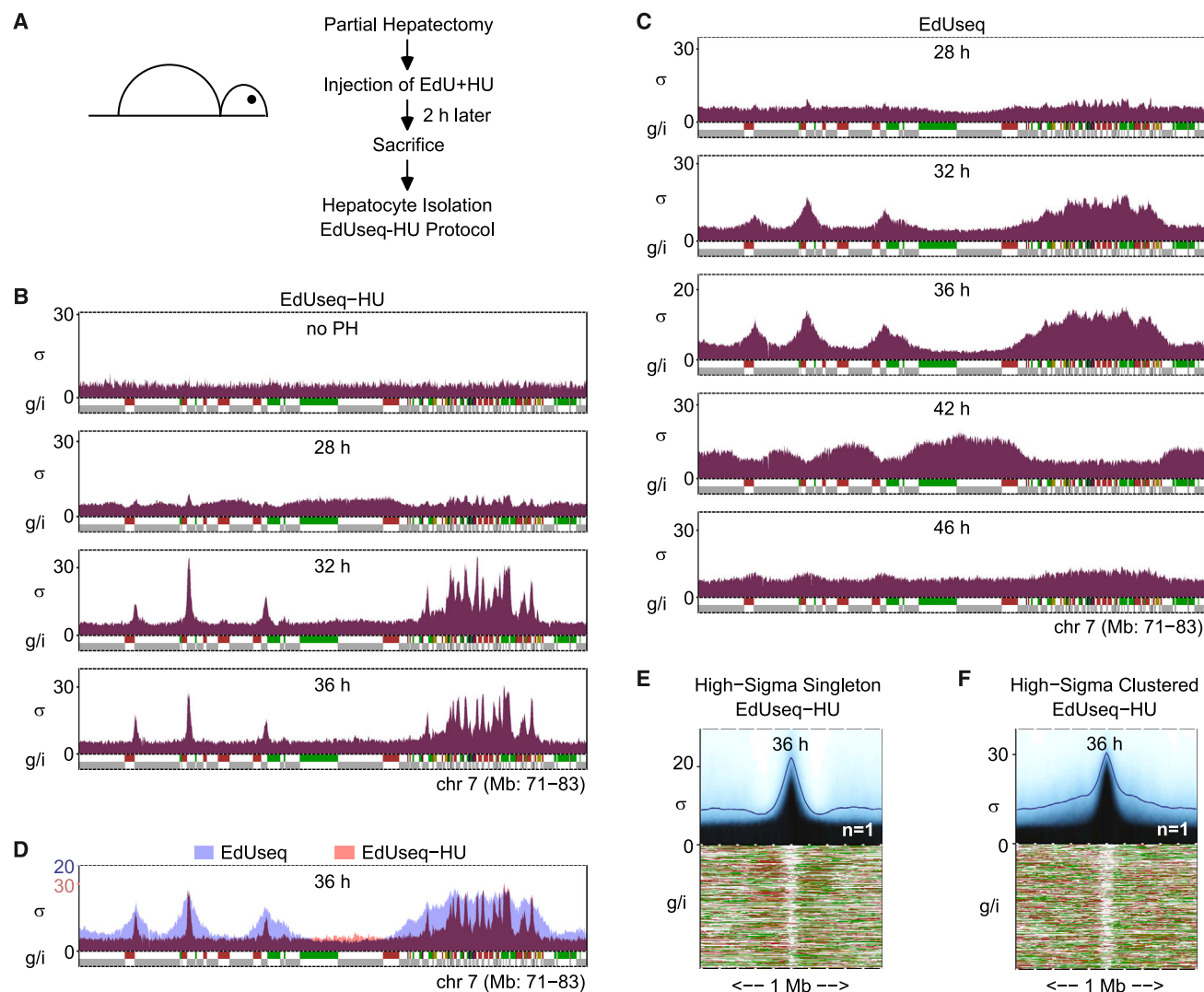


Figure 1. Firing of DNA replication origins in hepatocytes of adult young mice after partial hepatectomy

(A) Protocol for monitoring firing of DNA replication origins *in vivo*. HU, hydroxyurea.

(B) EdU incorporation profiles at a representative region of chromosome (chr) 7 (Mb: 71–83) in mice co-injected with EdU and HU. The livers were harvested at the indicated times post-partial hepatectomy (post-PH). σ , sigma values of EdU incorporation peaks; g/i, gene/intergenic regions. Genes are colored green or red for forward or reverse direction of transcription, respectively; intergenic regions are gray. Bin resolution, 10 kb; ruler scale, 100 kb.

(C) Profiles of *in vivo* EdU incorporation at the indicated times post-PH in mice treated as in (A), except that HU was not administered.

(D) Overlay of the EdU incorporation profiles at 36 h post-PH in mice treated with or without HU.

(E) Overlay plot of 36 h post-PH EdU-seq-HU signals at the 341 high-sigma singleton origins. The line indicates the average sigma value. The heatmap below the EdU-seq-HU plot shows the annotated protein-coding genes, colored according to the direction of transcription: green, forward; red, reverse; white, intergenic regions. The genomic regions in the heatmap were plotted from top to bottom in order of decreasing EdU-seq-HU peak sigma value. Span of genomic regions shown: 1 Mb; bin resolution: 10 kb; ruler scale: 100 kb; n, number of mice from which the data were obtained.

(F) As (E), except that data for the 349 high-sigma clustered origins were plotted.

See also Figure S1.

is incorporated into nascent DNA, whereas HU, a ribonucleotide reductase inhibitor, limits fork progression (Table S1A). The mice were sacrificed 2 h later, the hepatocytes were isolated, and the DNA that had incorporated EdU was subjected to high-throughput sequencing to identify DNA replication origins (Figure 1A). Examination of a representative region of chromosome (chr) 7 revealed no EdU incorporation after a sham PH, weak

EdU incorporation 28 h post-PH, and strong EdU incorporation 32 and 36 h post-PH (Figure 1B). These kinetics match those of hepatocyte cell-cycle entry after PH, as determined by immunohistochemistry (IHC).^{31–33}

To examine if the EdU incorporation peaks correspond to replication origins, we repeated the experiment without HU, so that fork progression would not be inhibited (Table S1B). At

32–36 h post-PH, there were broad zones of EdU incorporation (Figure 1C), whose centers corresponded to the EdU incorporation peaks observed in the presence of HU (Figure 1D). At 42 h post-PH, EdU was incorporated in the genomic domains that had not incorporated EdU at the earlier time points, consistent with replication of late S genomic domains. Finally, at 46 h post-PH, EdU incorporation marked again the early-replicating genomic domains, indicating the beginning of a second round of DNA replication (Figure 1C). Based on the above profiles, we conclude that the EdU-seq-HU method identifies origin firing sites *in vivo*; moreover, it appears to have very low false discovery rates, as we observed only the background signal in mice subjected to sham PH (Figure 1B).

To make a list of the origin firing sites, we identified three EdU-seq-HU datasets with the highest signal to noise ratios; two of these datasets were from mice sacrificed 32 h post-PH and one from a mouse sacrificed 36 h post-PH. The three datasets were merged and analyzed by a peak-finding algorithm. We will refer to the identified peaks as origin firing sites or origins, but since the resolution of the EdU-seq-HU method is 10 kb, we recognize that these sites may be replication initiation zones containing more than one origin. In total, 3,517 origin firing sites were identified, of which 920 were singletons, meaning that the signal reached background levels on both sides of the peak; the remaining peaks were in clusters of 2–29 peaks (Table S2A; Figure S1A, top). In the representative region of chr 7, the peaks on the left are examples of singletons and the peaks on the right examples of clusters (Figure 1B).

Next, we explored the variability in origin firing by comparing origin firing data from four mice sacrificed 32 h post-PH and one mouse sacrificed 36 h post-PH. For each mouse, we determined the sigma value (peak height divided by its standard deviation) for each origin. Then, we compared the sigma values of all 3,517 origins between pairs of mice by linear regression. The sigma values correlated very well between all pairs of mice studied, indicating that there is little variability in origin firing between mice (Figure S1B).

Of the 920 singleton peaks, we assigned 341 peaks with sigma values greater than 15 and no other origin within 200 kb to the high-sigma singleton group (Table S2B; Figure S1A, middle). Similarly, of the 2,597 clustered peaks, 349 peaks with sigma values greater than 25 were assigned to the high-sigma clustered group (Table S2C; Figure S1A, bottom). The assignment of origins into high-sigma singleton and high-sigma clustered groups allowed us to calculate average signals for the most efficient origins (Figures 1E and 1F). Average signals were also calculated for all 3,517 origins (Figure S1C). These averages confirmed the kinetics of origin firing that we observed at the representative region of chr 7 (Figures 1E, 1F, and S1C–S1E).

Nascent transcription in mouse regenerating livers

The location and usage of DNA replication origins are influenced by transcription.^{19,24} Thus, to study the coordination of DNA replication and transcription *in vivo*, we monitored nascent transcription in regenerating mouse livers. We focused initially on the 36-h time point post-PH, which corresponds to the peak of origin firing activity. Mice were injected with 5-ethynyl-uridine (EU), an analog of uridine, 35.5 h post-PH and sacrificed 30 min later;

their hepatocytes were isolated, and the EU-labeled nascent RNA was subjected to high-throughput sequencing (EU-seq; Table S1C). The EU-seq data were processed at a resolution of 10 kb to match the resolution of the origin firing data. Inspection of the representative region of chr 7 revealed that the nascent transcription signal, where present, coincided very well with the annotated genes, whereas the intergenic (ig) regions were devoid of signal (Figure S2A).

For each gene, we calculated its average log₂ EU-seq sigma value over all its genomic bins. A histogram plot of these values suggested that a value of 0.5 was a suitable threshold to distinguish expressed from non-expressed genes (Figure S2B). Reassuringly, the directionality of nascent transcription of the expressed genes, as determined by EU-seq, matched the gene annotation data very well (Figure S2C).

The majority of expressed genes are replicated in early S phase

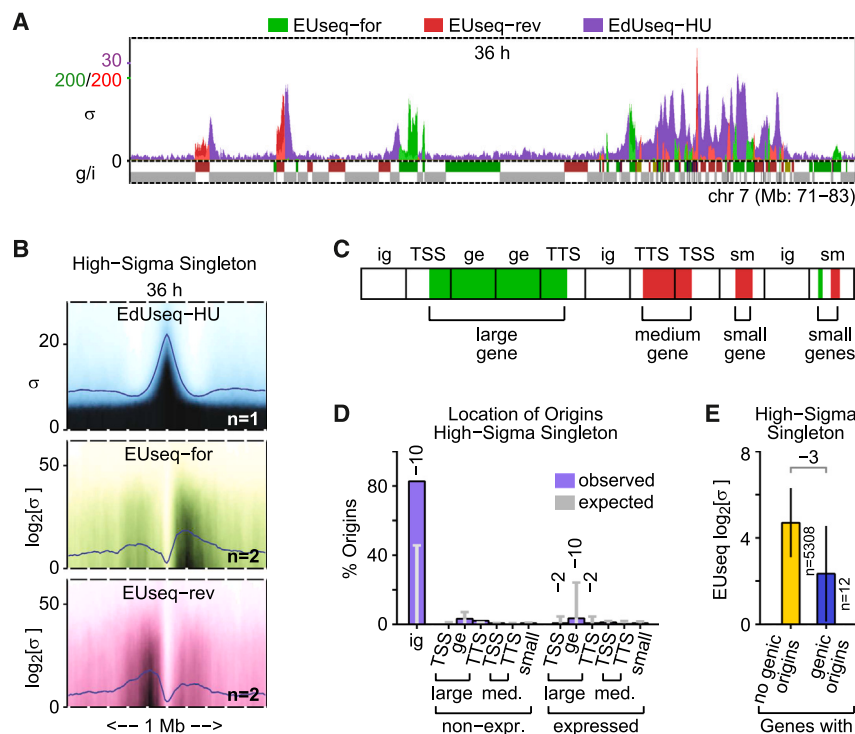
In tissue culture cells, expressed genes are replicated preferentially in early S phase.^{34–39} We used the *in vivo* EdU-seq incorporation data at 32 and 42 h post-PH (Figure 1C) to define early- and late-replicating genomic domains. About half the mouse genome was early replicating and the other half was late replicating (Figure S2D). The non-expressed genes were distributed equally in the early- and late-replicating genomic domains; however, the expressed genes were preferentially localized in the early-replicating genomic domains (Figure S2E). Moreover, the early-replicating expressed genes had almost 2-fold higher levels of nascent transcription than the late-replicating expressed genes (Figure S2F).

Origin firing sites map preferentially to non-transcribed genomic regions

Next, we superimposed the origin firing and nascent transcription data collected 36 h post-PH. Within the representative region of chr 7, the origins fired near, but not within, transcribed regions (Figure 2A). This was particularly evident in genome-wide plots, as there was a strong nascent transcription signal on either side of DNA replication origins but practically no transcription signal at the origin firing sites (Figures 2B and S2G).

To further study the genomic location of origin firing sites, the gene annotation data were used to assign labels to each 10-kb genomic bin. Bins containing only genic (ge) or only ig sequences were assigned ge and ig labels, respectively. The remaining bins, containing both ig and ge sequences, were assigned transcription start site (TSS), transcription termination site (TTS) or small gene (sm) labels. The TSS and TTS labels were reserved for medium and large genes, defined as genes extending over two or more than two genomic bins, respectively. The sm label was reserved for genes that mapped within a single bin (Figure 2C).

The majority of origin firing sites mapped to ig bins and did so at a frequency that significantly exceeded the frequency of ig bins in the early-replicating domains of the genome (Figures 2D and S2H). Reciprocally, origin firing within the bodies of expressed genes was strongly disfavored, and the very small number of expressed genes that harbored replication origins within their gene bodies was weakly expressed (Figures 2D, 2E, S2H,



(E) Nascent transcription of large, early-replicating expressed genes with no origins or high-sigma singleton origins within their gene bodies. For each group of genes, the mean EU-seq log₂ sigma value and its standard deviation are shown. The level of statistical significance is indicated by the log₁₀ of the *p* value; *n*, number of genes.

See also Figure S2.

and S2I). For all other bin types, origin firing was neither favored nor disfavored (Figures 2D and S2H).

Proximity of ig origin firing sites to TSS or TTS

The observation that origins fire near expressed genes (Figures 2B and S2G) has implications for co-directionality of transcription and replication. For each early-replicating expressed gene, we scored for the presence of ig origins located within 100 kb of the gene transcribed sequence. One bioinformatics search started from the TSS, another from the TTS; the searches stopped when either an origin or another expressed gene was identified or when the 100-kb threshold was reached; the selected 100-kb threshold corresponds to the average inter-origin distance in human cells.⁴⁰

The transcribed sequences of about half of all the early-replicating expressed genes were near an origin, with most of these genes being near only one origin (Figure 3A). For a little more than half of these genes, the origin in proximity was upstream of the TSS, suggesting that these genes were replicated co-directionally with transcription, whereas the rest of the genes were likely to be replicated head-on with transcription, since the proximal origin was downstream of the TTS (Figures 3A and S3A). The average distance of the proximal origin to the transcribed sequence was similar for the genes replicated co-directionally and head-on (Figure S3B); however, the average nascent transcription signal was about 2-fold higher in the former than the latter genes (Figure S3C). Of

Figure 2. DNA replication origins map to intergenic regions

(A) EdU and EU *in vivo* incorporation profiles at the representative region of chr 7 (Mb: 71–83). All mice were sacrificed 36 h post-PH. The EdU-seq-HU profile is from the experiment of Figure 1B. The nascent transcripts are colored green and red to indicate forward (for) and reverse (rev) transcription, respectively. The data are plotted as in Figure 1B. (B) Overlay plots of 36-h post-PH EdU-seq-HU and EU-seq (forward and reverse directionality) signals at the 341 high-sigma singleton origins, plotted as in Figure 1E. The EdU-seq-HU data are reproduced from Figure 1E. *n*, number of mice from which the data were obtained.

(C) Scheme by which the 10-kb genomic bins were classified according to gene annotation. Black vertical lines mark the genomic bins; filled boxes indicate hypothetical genes colored green or red for forward and reverse transcription, respectively. Classification codes: ig, intergenic; ge, genic; TSS, transcription start site; TTS, transcription termination site; sm, small gene(s).

(D) Distribution of the high-sigma singleton origins on genomic bins, classified according to the scheme of (C). The expected frequencies refer to the fraction of each bin type in the early-replicating domains of the mouse genome. For bin types with statistically different observed and expected frequencies, the log₁₀ of the *p* value is shown.

the subset of genes replicated by high-sigma origins, close to two-thirds were replicated co-directionally with transcription (Figures S3A–S3C). These results suggest a preference for co-directionality of replication and transcription, despite many transcribed genes being replicated head-on with transcription.

To further study the proximity of origins to expressed genes, we plotted the distance of origins from the TSS or TTS, extending the search up to 300 kb away from the TSS or TTS (Figure 3B). The analysis took into account the presence of neighboring expressed genes, and the search for origins stopped when such a gene was identified. Interestingly, there was a strong distance dependence for origin firing. The probability of origins firing within bins 2–5, which correspond to distances of 10–50 kb from the TSS or TTS, was statistically significantly greater than expected, whereas the probability of origin firing further away was lower than expected (Figure 3B). This finding raises the question whether, from an evolutionary perspective, the position of origins on the genome is selected to be near transcribed sequences.

Evolutionary conservation of origin firing sites

To determine whether the position of origins relative to transcribed sequences is conserved in evolution, we studied origin firing in human cells. We selected the mouse genes that were expressed in regenerating livers and that had only one proximal origin of replication; these genes were split into TSS and TTS

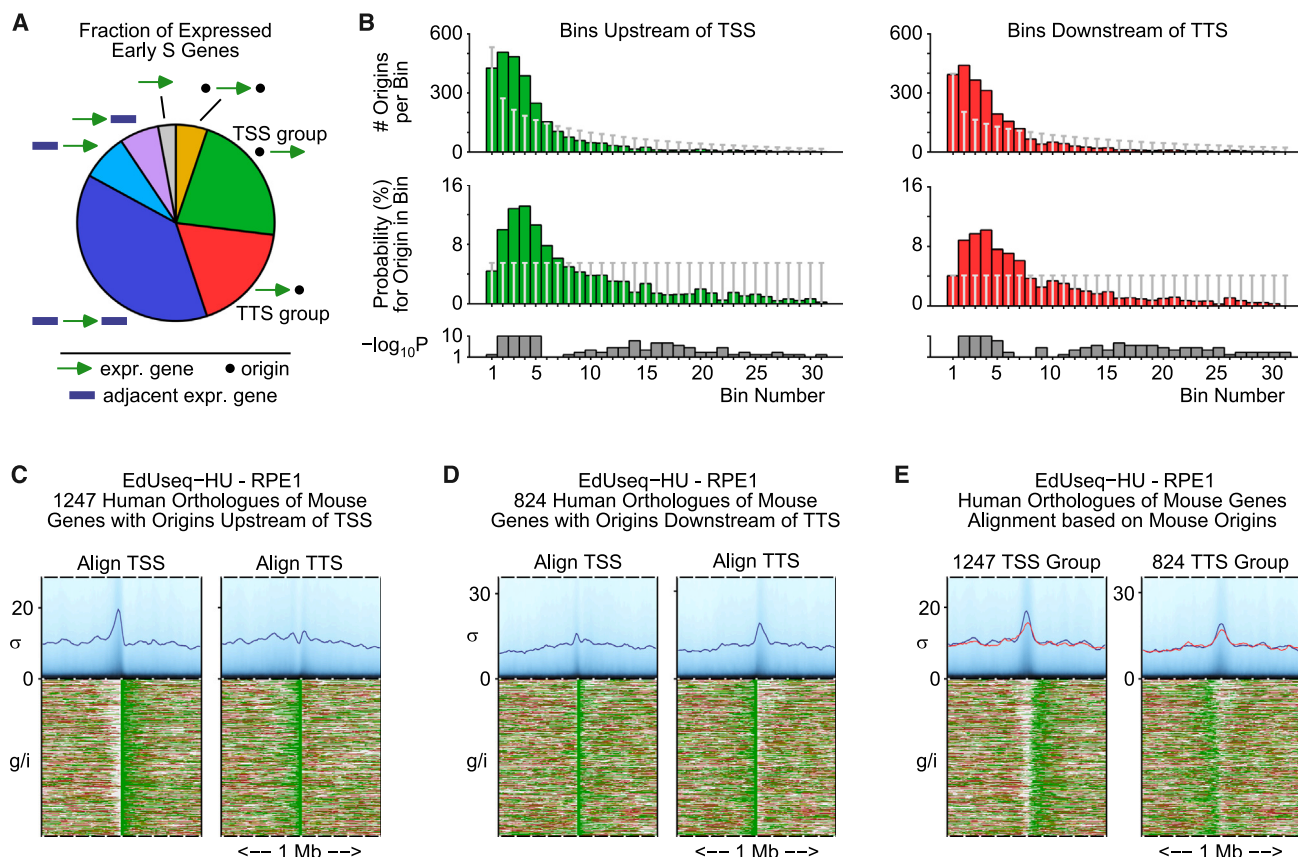


Figure 3. Proximity and conservation of position of origins to expressed genes

(A) Distribution of expressed, early-replicating genes according to the presence of a replication origin or other expressed gene in closest proximity to their transcription start site (TSS) and transcription termination site (TTS).

(B) Distribution of origins as a function of distance from the TSS and TTS. Observed and expected number of origins and probability that an origin will be present at a specific bin are shown. The calculation of the expected number of origins per bin took into account the decrease in the number of intergenic bins as the distance from the TSS or TTS increases, due to the presence of adjacent expressed genes. The statistical significance of the differences between observed and expected frequencies is shown as $-\log_{10}$ of p values corrected for multiple comparisons.

(C) Overlay plots of EdU-seq-HU signals at the 1,247 human genes that are orthologs of the TSS group of mouse genes (as defined in A). The plots were obtained by aligning the TSS (left) or TTS (right) of the human genes. The data were obtained from RPE1 cells and are plotted as in Figure 1E. The heatmap below the EdU-seq-HU plot shows the annotated protein-coding genes for only 300 of the 1,247 genomic regions.

(D) Overlay plots of EdU-seq-HU signals at the 824 human genes that are orthologs of the TTS group of mouse genes (as defined in A). Otherwise, the data were obtained and plotted as in (C).

(E) Overlay plots of EdU-seq-HU signals at the 1,247 (left) and 824 (right) human genes that are orthologs, respectively, of the TSS and TTS groups of mouse genes (as defined in A). The alignments took into account the distance of the mouse origin from the TSS (left) or TTS (right); otherwise, the data were obtained and plotted as in (C). The red line indicates the average sigma value for an alignment, for which the distance of the mouse origin from the TSS or TTS was randomly permuted. See also Figure S3.

groups, depending on whether the proximal origin was upstream or downstream of the transcribed sequence (Figure 3A). Next, we identified the human orthologs of the mouse TSS-group and TTS-group genes, aligned them by their TSS or TTS, and plotted the origin firing signal, obtained from human immortalized retinal pigment epithelial 1 (RPE1) cells. Interestingly, the human orthologs of the mouse TSS-group genes showed stronger than expected EdU incorporation signal upstream of their transcribed sequence (Figure 3C; $p < 10^{-5}$), whereas the human orthologs of the mouse TTS-group genes showed stronger than expected signal downstream of their transcribed sequence (Figure 3D; $p < 10^{-5}$).

Next, we examined if the distance between origins and the TSS or TTS was conserved from mouse to human. We aligned the human TSS-group and TTS-group orthologs on the basis of the origin firing sites of the corresponding mouse genes and then plotted the origin firing signal. This alignment led to a robust EdU incorporation signal (Figure 3E) that was stronger than any signal that could be obtained after randomly permuting the mouse origin-TSS and origin-TTS distances for alignment of the human orthologs (Figure 3E; $p < 10^{-4}$). We conclude that not only the position of the origin, upstream or downstream of the transcribed sequence, but also the distance of the origin from the TSS or TTS are conserved in evolution.

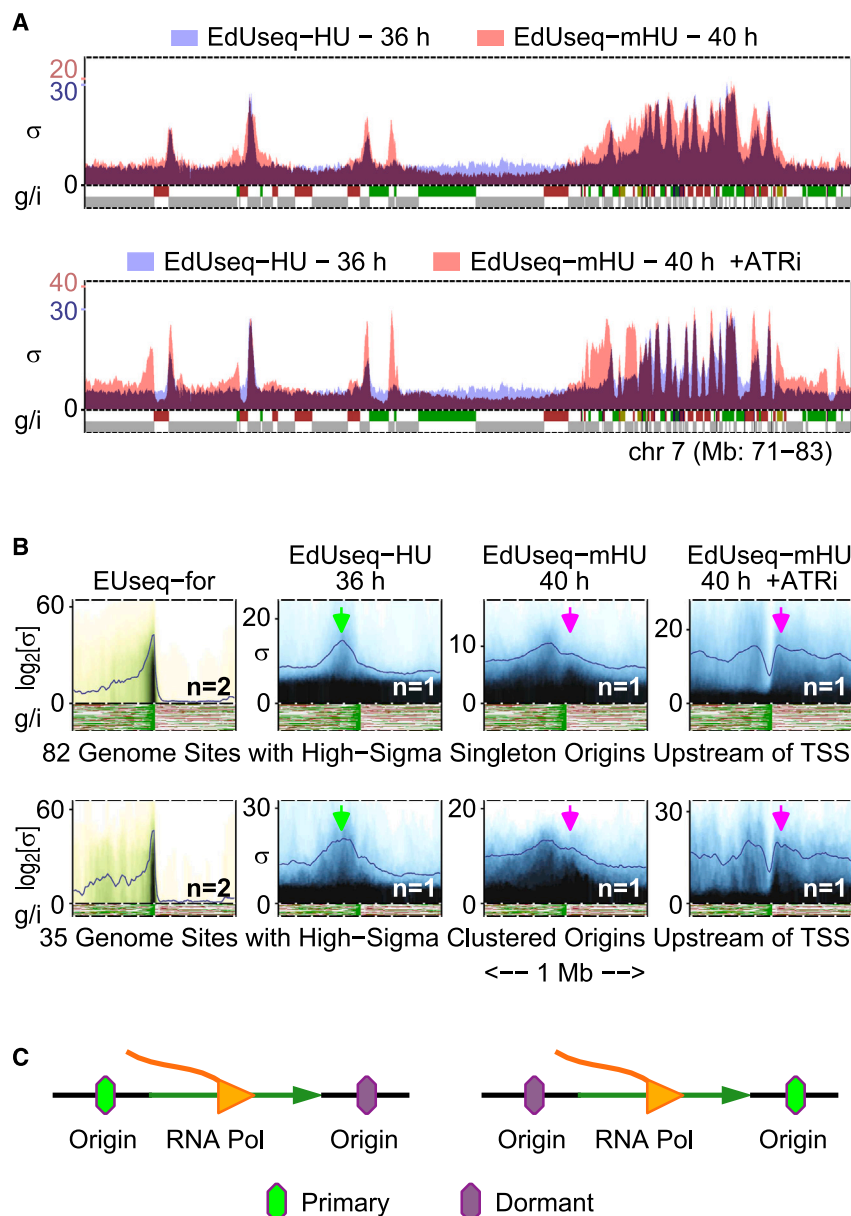


Figure 4. Dormant origin firing

(A) EdU *in vivo* incorporation profiles at the representative region of chromosome 7 (Mb: 71–83). The mice sacrificed 40 h post-PH had HU administered at 34, 36, and 38 h post-PH (multiple HU [mHU]) to elicit DNA replication stress; one of these mice was also injected with an ATR inhibitor (ATRi) 24 h post-PH. The 36-h post-PH EdU-seq-HU profile is from Figure 1B, and the data are plotted as in Figure 1B. (B) Overlay plots of EU-seq and EdU-seq-HU signals at genomic regions containing genes with high-sigma singleton (top) or high-sigma clustered (bottom) origins upstream of their TSS and no origin or expressed gene within 400 kb downstream of their TTS. The genes are plotted so that the direction of transcription is from left to right, and they are aligned by their TTS. The 36-h post-PH EU-seq and EdU-seq-HU datasets are from the same mice as in Figure 2B; the 40-h post-PH EdU-seq-HU datasets are from the mice of (A). The green and purple arrows indicate the primary origins upstream of the TSSs and the dormant origins downstream of the TSSs, respectively. The data are plotted as in Figure 1E.

(C) Cartoon showing genes (green arrows) with two replication origins, one primary and one dormant, flanking the transcribed sequence. See also Figure S4.

Inspection of the representative region of chr 7 revealed dormant origin firing in mice experiencing HU-induced DNA replication stress, particularly when ATR was also inhibited (Figure 4A). For a genome-wide analysis, we identified the expressed genes that had on one side a primary origin and on the other side, over a distance of 400 kb, neither a primary origin nor an expressed gene. The 400-kb span allowed us to monitor dormant origin firing without influence from nearby expressed genes. The genes with primary origins upstream of the TSS were aligned by their TTS, whereas the genes with primary origins downstream of the TTS were aligned by their TSS. These alignments revealed

firing of dormant origins that, similar to primary origins, were located near the expressed sequences (Figures 4B and S4) but on the opposite side of the gene than the primary origins (Figure 4C).

Poor origin firing efficiency in aged mice after PH

The fraction of hepatocytes entering S phase after PH decreases with age.^{49,50} We examined origin firing in regenerating livers of 12-, 18- and 22-month-old mice using the same protocol, as for young mice. Sixteen aged mice were examined. In two mice, both from the 12-month-old cohort, origin firing was as efficient as in young mice; however, in all other mice, origin firing was inefficient, as seen in the representative examples of two 18-month-old mice (Figure 5A). Genome-wide

Firing of dormant DNA replication origins *in vivo*

In tissue culture cells, DNA replication stress can induce firing of dormant replication origins.^{24,41–44} Since we were able to map origins, hereafter referred to as primary origins, in otherwise untreated mice post-PH, we wondered whether agents that induce DNA replication stress would lead to firing of additional, dormant origins. After PH, mice received HU three times, at 34, 36, and 38 h post-PH, to block fork progression over an extended period of time; EdU was administered at the 38-h time point, and the hepatocytes were isolated 2 h later. In a second regimen, the mice also received an inhibitor of the ATR kinase 24 h post-PH. ATR is a key player in the response to DNA replication stress,^{45,46} and its activity can be inhibited by small chemical compounds *in vitro* and *in vivo*.^{47,48}

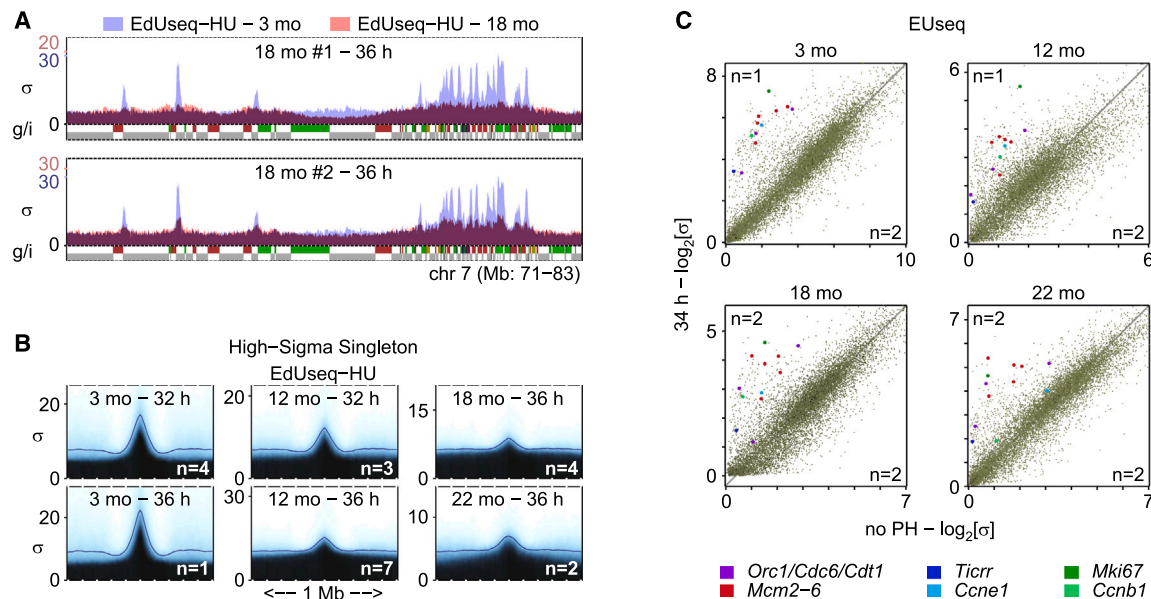


Figure 5. DNA replication origin firing defect in hepatocytes of aged mice

(A) Comparison of 36-h post-PH EdU-seq-HU profiles of two 18-month-old mice (#1 and #2) and a 3-month-old mouse. The mice were treated as in Figure 1A. A representative region of chr 7 (Mb: 71–83) is shown. The profile of the young mouse is from Figure 1B, and the data are plotted as in Figure 1B. mo, month.
 (B) Overlay plots of EdU-seq-HU signals at the 341 high-sigma singleton origins in mice of the indicated ages 32 or 36 h post-PH. The data are plotted as in Figure 1E; *n*, number of mice from which the data were obtained.
 (C) Induction of gene expression in response to PH, monitored by EU-seq, before and 34 h post-PH in 3-, 12-, 18- and 22-month-old mice. Each data point corresponds to one gene; the data points corresponding to select genes involved in DNA replication are color coded, as shown. σ , sigma value; *n*, number of mice from which the data were obtained.

See also Figure S5.

averages of all origins, or of the high-sigma origins, confirmed the aging-associated defect in origin firing (Figures 5B and S5A).

We wondered if the origin firing defect was accompanied by poor induction of genes required for DNA replication. Young and old mice were administered EU either at various time points after PH or after a sham PH; 30 min later the mice were sacrificed, and the hepatocytes were isolated to monitor nascent transcription *in vivo*. In young mice, we observed induction of expression of genes required for origin licensing and firing (*Orc1*, *Cdc6*, *Cdt1*, and *Tcrr*), replisome progression (*Mcm2–6*), and for cell-cycle progression (*Ccne1*, *Ccnb1*, and *Mki67*) 28–34 h post-PH, coinciding with entry into S phase (Figures 5C and S5B). Expression of the same genes was also induced in the old mice (Figures 5C and S5C), suggesting that the origin firing defect was not due to poor gene expression. In a previous study monitoring gene expression by microarrays after PH, the cell-cycle-regulated genes were also induced equally well in young and old mice.⁵¹

The replication stress checkpoint suppresses origin firing in aged mice

The ATR checkpoint kinase suppresses origin firing in cells that enter S phase under conditions of DNA replication stress.^{43,52–59} To illustrate this effect, we incubated U2OS osteosarcoma cells with the DNA polymerase inhibitor aphidicolin and monitored EdU incorporation as the cells entered S

phase. Aphidicolin suppressed EdU incorporation but not when the cells were co-treated with an ATR inhibitor (Figure S6A). To replicate this experiment *in vivo*, we administered an ATR inhibitor 24 h post-PH to aged mice and monitored origin firing at 36 h. We observed very efficient origin firing in all the 15 aged mice treated with the ATR inhibitor, including the 22-month-old mice (Figures 6A, 6B, and S6B). Pairwise comparisons of the sigma values of origin peaks from young and old mice revealed that firing of all origins was rescued in the old mice treated with the ATR inhibitor (Figure S6C). In young mice, the ATR inhibitor had no effect, as origin firing was already efficient (Figures 6A, 6B, S6B, and S6C). Finally, inhibiting ataxia telangiectasia mutated (ATM), an ATR-related checkpoint kinase activated by DNA double-strand breaks,¹¹ did not rescue origin firing in the old mice (Figures 6B and S6B).

The effect of the ATR inhibitor in aged mice raised the question whether PH was accompanied by induction of a DNA damage response. We monitored the levels of the DNA damage marker γ H2AX in hepatocytes isolated from young and aged mice after PH or after sham PH. Initially, 2 h prior to being sacrificed, the mice were injected with EdU. In young mice (*n* = 11), PH led to a significant increase in the fraction of EdU-positive hepatocytes; the percentage of γ H2AX-positive cells also increased but only mildly. The inverse pattern was observed in aged mice (*n* = 9); PH led to a small increase in the fraction of EdU-positive cells and a strong increase in the fraction of γ H2AX-positive cells

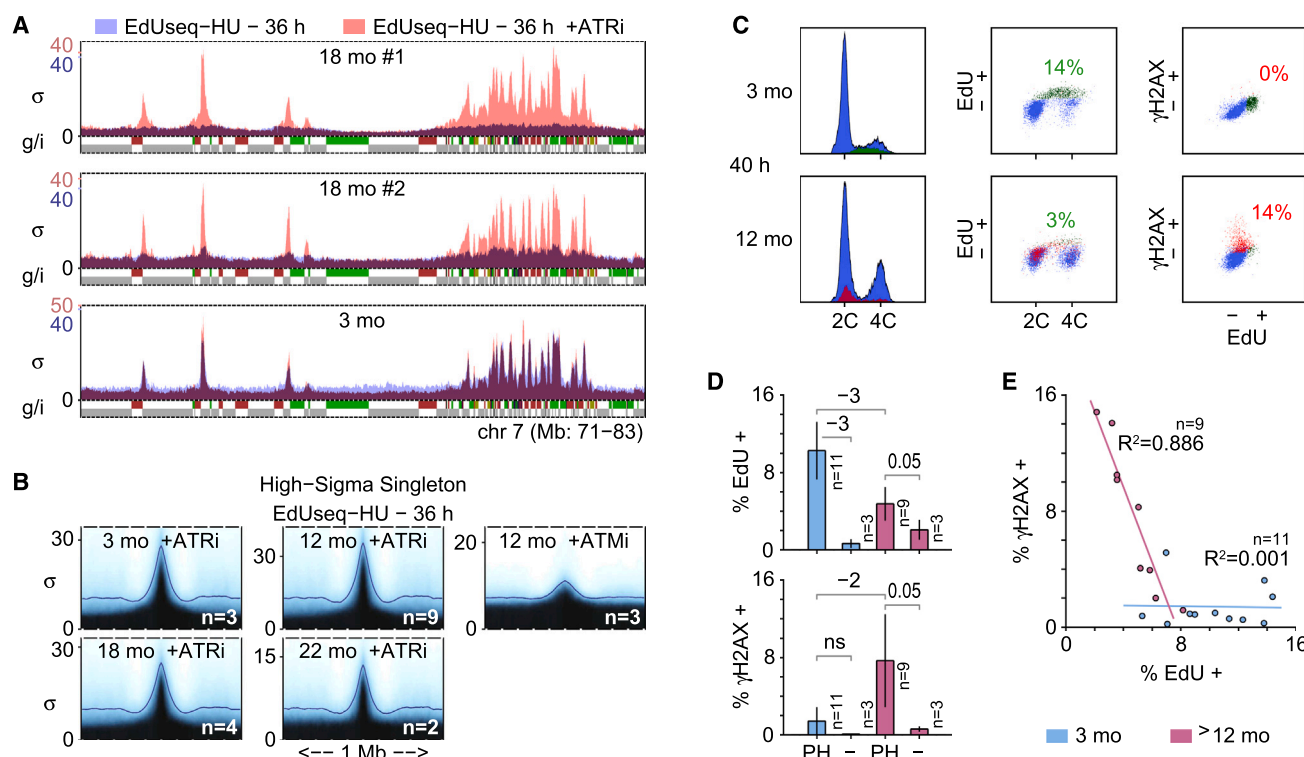


Figure 6. DNA replication stress underlies the origin firing defect in hepatocytes of aged mice

(A) Increased origin firing efficiency in 18-month-old mice treated with an ATR inhibitor (ATRi) 24 h post-PH. EdU-seq-HU profiles of hepatocytes of 18-month-old mice (mice #1 and #2, from Figure 5A) are compared with the profiles of two 18-month old mice treated with ATRi; the profile of a 3-month-old mouse (Figure 1B) is also compared with the profile of a young mouse treated with ATRi. A representative region of chromosome 7 (Mb: 71–83) is shown; the data are plotted as in Figure 1B. mo, month.

(B) Overlay plots of EdU-seq-HU signals at the 341 high-sigma singleton origins 36 h post-PH in mice injected with ATRi or ATM inhibitor (ATMi). The data are plotted as in Figure 1E; n, number of mice from which the data were obtained.

(C) Representative flow cytometry data of DNA content, EdU incorporation, and γ H2AX levels in hepatocytes isolated from 3- and 12-month-old mice 40 h post-PH. The mice were injected with EdU, 2 h before being sacrificed. The percentages of EdU-positive (green) and γ H2AX-positive (red) cells are shown. 30,000–50,000 cells were analyzed per sample.

(D) Fraction of EdU-positive (% EdU+) and γ H2AX-positive (% γ H2AX+) hepatocytes in livers of untreated mice (–) or 40 h post-PH (PH). The mice were injected with EdU, 2 h before being sacrificed. Means, standard deviations, and number of mice (n) are shown; the level of statistical significance is indicated as $\log_{10}p$ units, except for $p < 0.05$, which is indicated as such; ns, not significant.

(E) Inverse correlation between the fraction of hepatocytes entering S phase and the fraction exhibiting a DNA damage response in old, but not young, mice 40 h post-PH. The graph was generated using the data of (D).

See also Figure S6.

(Figures 6C and 6D). Moreover, there was a statistically significant negative correlation between EdU incorporation and γ H2AX expression (Figure 6E). The induction of a DNA damage response in aged mice after PH was further confirmed by IHC analysis of sections of liver lobes obtained at the time of PH and 48 h post-PH (Figure S6D).

Bypassing the replication stress checkpoint in aged mice leads to an inflammatory response

The rescue of the origin firing defect by the ATR inhibitor in aged mice (Figure S6C) prompted us to examine whether other aspects of liver regeneration were also rescued. At 48 h post-PH, almost 60% of the hepatocytes of young mice were positive for the Ki67 proliferation marker. In the 12-month-old mice, the corresponding percentage was about 30%. Treatment with the ATR inhibitor led to an increase in the fraction of Ki67-positive

hepatocytes to a level of about 40% (Figure 7A). However, this increase was not statistically significant.

Next, we monitored changes in gene expression 48 and 72 h post-PH in young and old mice treated with or without ATR inhibitor. Steady-state levels of mRNAs isolated from whole livers were assessed by RNA-seq, and for each condition, datasets from three or four mice were combined to reduce variance. In the absence of ATR inhibitor, the gene expression patterns after PH, in both the young and old mice, revealed increased expression of genes related to cell-cycle progression (Figures 7B–7D and S7B; Table S3). Administering ATR inhibitor had no further effect on gene expression in the young mice (Figures 7D and S7A), whereas in the old mice, the ATR inhibitor led to increased expression of genes linked to immune responses and inflammation, specifically at the later time point of 72 h post-PH (Figures 7B–7D and S7A–S7C; Table S4).

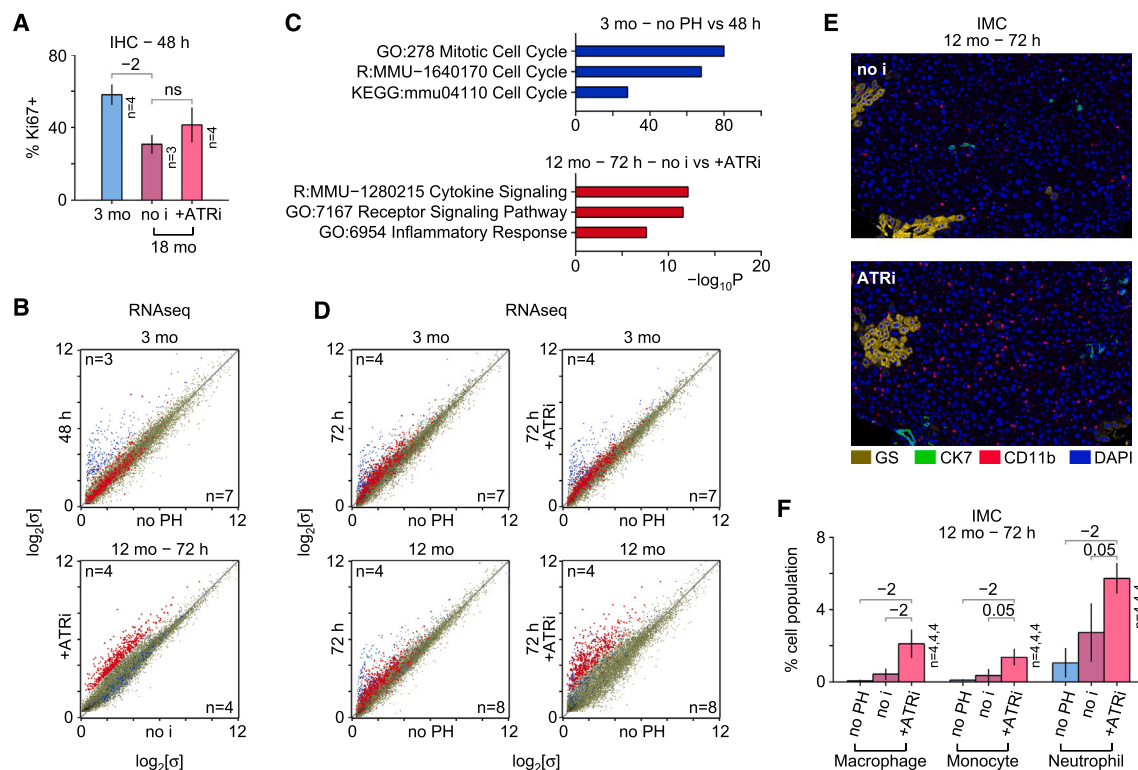


Figure 7. Inflammatory response 72 h post-PH in aged mice treated with an ATR inhibitor

(A) Fraction of hepatocytes expressing the Ki67 proliferation marker (% Ki67+) as assessed by immunohistochemistry (IHC) 48 h post-PH. Old mice were untreated (no i) or treated with an ATR inhibitor (+ATRi); the young mice did not receive an ATR inhibitor. Means, standard deviations, and number of mice (*n*) are shown; the level of statistical significance is indicated as log₁₀*p* units; ns, not significant; mo, month.

(B) Top: plots of gene expression values before and 48 h post-PH in 3-month-old mice; genes whose expression was induced by more than three standard deviations in response to PH (data points colored blue; Table S3) were used to identify gene signatures. Bottom: plots of gene expression values 72 h post-PH in 12-month-old mice treated with or without an ATR inhibitor; genes induced by more than three standard deviations by the ATR inhibitor (data points colored red; Table S4) were used to identify gene signatures. Each data point corresponds to one gene and is the average of the values from *n* mice. σ, sigma value.

(C) Gene signatures associated with entry into the cell cycle after PH in young mice and with an inflammatory response in response to treatment with the ATR inhibitor in aged mice.

(D) Induction of expression of cell-cycle genes in young and aged mice 72 h post-PH and induction of expression of inflammatory response genes in the aged mice that received ATR inhibitor. The data are plotted and color coded as in (B).

(E) Imaging mass cytometry (IMC) of liver sections 72 h post-PH from aged mice treated without (no i) or with ATR inhibitor (ATRi). GS (Glu), glutamate-ammonia ligase; CK7 (KRT7), keratin 7; CD11b (Itgam), integrin alpha M.

(F) Fraction of macrophages, monocytes, and neutrophils in liver sections of aged mice, as determined by IMC. Liver tissues were obtained at the time of PH (no PH) or 72 h post-PH (no i, no inhibitor; ATRi, ATR inhibitor). Means, standard deviations, and number of mice in each group (*n*) are shown; the level of statistical significance is indicated as log₁₀*p* units, except for *p* < 0.05, which is indicated as such.

See also Figure S7.

To understand the nature of this inflammatory response, we performed imaging mass cytometry (IMC) on histology sections from old mice 72 h post-PH. The liver lobes resected during the PH procedure served as no PH controls. Mice treated with or without ATR inhibitor were studied. The fraction of resident liver cells, including hepatocytes and Kupffer cells, and the fraction of B and T cells were not affected by PH or the ATR inhibitor treatment (Figure S7D). In contrast, the fraction of infiltrating macrophages, monocytes, and neutrophils increased significantly 72 h post-PH in the mice treated with the ATR inhibitor (Figures 7E and 7F). Thus, the rescue of the origin firing defect in aged mice at early time points post-PH by the ATR inhibitor was accompanied by induction of an inflammatory response at the later time points.

DISCUSSION

Mapping origin firing sites *in vivo*

Mapping DNA replication origins on mammalian genomes is challenging. We overcame the technical difficulties to accomplish this task *in vivo* by studying regenerating mouse livers. We obtained EdU incorporation profiles at various time points after PH from mice treated with or without HU, as well as from mice subjected to sham PH. The collected data convincingly support the premise that the peaks identified by the EdU-seq-HU method correspond to genuine origins, whose genomic position we used to compile a list of mammalian origin firing sites.

In tissue culture cells, two distinct subsets of replication origins have been identified. The first subset includes origins

identified by the OK-seq, EdU-seq-HU, and REPLI-seq methods.^{19,24–27} Many origins in this subset are *ig*, located at TSSs or up to 20 kb upstream of the TSSs and are conserved in evolution.^{19,24,60} The second subset contains the core, consensus origins identified by the SNS-seq and INI-seq methods, most of which are located within genes and are not evolutionarily conserved.^{20,21,28–30} The *in vivo* origin firing sites shared features with the origins of the first subset; i.e., they were *ig*, located near the TSS or TTS of expressed genes, and their genomic positions were conserved in evolution.^{19,24,60}

One notable difference between the origin firing sites identified by EdU-seq-HU *in vivo* and OK-seq in tissue culture cells is how big a region they span on the genome. The origin firing sites identified by OK-seq *in vitro* are broad initiation zones, up to 75 kb wide, and are thought to contain multiple origins, only one of which fires stochastically at each cell cycle.¹⁹ Instead, the origin firing sites identified by EdU-seq-HU *in vivo* were mostly contained within 10-kb genomic bins. The different length spans could be due to differences in resolution of the EdU-seq-HU and OK-seq methods, although we cannot rule out the possibility that *in vivo* origin firing may be qualitatively and quantitatively different from *in vitro* firing, for example, less stochastic *in vivo* than *in vitro*.

Location of origins relative to transcription units

The replication initiation zones identified by OK-seq in tissue culture cells are located preferentially at promoters or within 20 kb of the promoters of expressed genes.^{19,24} The *in vivo* origin firing sites were also proximal to the transcribed sequences of expressed genes, but nearly as many origins were located downstream of the TTS, as upstream of the TSS. Moreover, the *in vivo* origins were located 10–50 kb away from the TSS or TTS (Figure 3B), whereas the *in vitro* studies identified many origins at gene promoters.^{19,24}

The location of origins relative to transcription units is likely to impact genome stability and replication timing. The rate of accumulation of genome rearrangements in normal somatic cells is very low.⁶¹ When genome rearrangements arise, transcription replication conflicts (TRCs) are thought to be a main driver.^{62–66} The propensity of a TRC to compromise genome stability depends on the type of TRC. The most mutagenic TRCs are those involving origins that fire within expressed genes.^{27,67,68} TRCs of this type can lead to genome rearrangements, as is evident in cancers, in which oncogenes induce aberrant origin firing within transcribed genes.^{27,67,68} *In vivo*, these mutagenic TRCs were avoided by locating almost all origins in *ig* regions (Figures 2D and 2E).

The second most mutagenic type of TRCs is head-on TRCs involving origins that fire downstream of gene TTS.^{62–66} These TRCs were disfavored by positioning the origins upstream of the TSSs, particularly for the highly transcribed genes (Figure S3A). Nevertheless, a significant fraction of expressed genes had origins downstream of the TTS. Some of these genes may contain replication termination sites at the TTS,^{24,69} but the rest would likely be replicated head-on with transcription, suggesting that head-on TRCs may not be particularly mutagenic *in vivo*. Contending with head-on TRCs provides flexibility in

genome organization and allows large genes to be replicated from two origins located at the upstream and downstream *ig* regions, respectively.⁷⁰

In regard to replication timing, the proximity of origins to TSSs or TTSs helps explain why expressed genes tend to be replicated in early S phase.^{34–39} The open chromatin state associated with expressed genes can encompass the nearby replication origins, favoring the efficient recruitment of origin licensing and firing factors. DNA replication error rates are lowest in early S phase.^{71,72} Thus, linking origin firing efficiency to transcriptional activity ensures that expressed genes are replicated under the most favorable conditions.

The DNA replication stress checkpoint curtails origin firing in aged mice

In old mice subjected to PH, there was a defect in origin firing that could be fully rescued by administering an ATR inhibitor (Figure 6A). These results suggest that the DNA replication stress checkpoint curtails origin firing in aged mice. Consistent with this, PH led to a DNA damage response, whose magnitude correlated inversely with the fraction of hepatocytes entering S phase (Figure 6E). By comparison, the hepatocytes of young mice exhibited a very mild DNA damage response after PH, and origin firing was robust and not further enhanced by inhibiting ATR.

DNA damage has been linked to aging.^{7–9} Several mechanisms have already been proposed. DNA damage may introduce detrimental mutations in essential genes,⁷³ or activate ATM and p53, leading to cell senescence^{74–76} or affecting the epigenetic state and gene expression.⁷⁷ One study demonstrated decreased levels of MCM2-7 helicase subunits in hematopoietic stem cells of aged mice, leading to DNA replication stress.⁷⁸ In our study, inhibiting ATM in the old mice did not enhance origin firing efficiency; induction of gene expression after PH was similar in young and old mice; and the expression levels of the *Mcm2-7* genes were not reduced in the livers of old mice, suggesting that hepatocytes are impacted by other aging-dependent mechanisms.

We hypothesize that during aging, cells accumulate DNA damage, which damage remains undetected, or cryptic, as long as the cells remain quiescent. Upon entry into S phase, this cryptic DNA damage may be detected, possibly as the DNA replication machinery unwinds the parental DNA strands, leading to DNA replication stress, activation of ATR, and suppression of further origin firing. The nature of the presumed cryptic DNA damage remains elusive; it might include DNA interstrand crosslinks (ICLs), aldehyde-DNA adducts, oxidized bases, and abasic sites, all of which have the potential to stall DNA replication.^{79–84} We note that impaired repair of DNA ICLs leads to premature aging,^{85–87} and some ICLs may remain undetected/cryptic, until either transcription or DNA replication attempt to unwind the DNA strands.^{88–90}

If aging is due to DNA replication stress induced by cryptic DNA damage, then we might explain the conundrum why slowly proliferating tissues, such as the liver, age faster than highly proliferating tissues like the intestine.^{91–94} In cells that have remained quiescent for long periods of time, enough cryptic DNA damage may have accumulated near origins to induce a robust

replication stress response upon origin firing. In contrast, in fast proliferating tissues, little damage can accumulate between consecutive cell cycles.

Our model further predicts that it is not the cryptic DNA damage itself that induces aging but the cellular response to this damage. The cryptic DNA damage that accumulates during aging may not affect the function and viability of quiescent cells. Rather, cell senescence may occur when cells try to replicate their DNA under conditions of DNA replication stress. If it were possible to repair the cryptic DNA damage before the cells entered S phase, then perhaps some aspects of aging could be prevented.

Interestingly, while inhibiting ATR fully rescued origin firing efficiency in old mice, it did not enhance the fraction of hepatocytes entering the cell cycle to the levels seen in young mice and, moreover, leading to liver inflammation (Figure 7). The latter observation is consistent with previous studies showing that inhibiting ATR under conditions of DNA replication stress leads to inflammation. Mechanistically, ATR inhibitors can induce cytoplasmic DNA that activates the cGAS-STING pathway, as well as affect nucleotide metabolism leading to incorporation of interferon-inducing deoxyuridine in genomic DNA, resulting in induction of an innate immune response.^{95–102} Importantly, our study comparing *in vivo* replication dynamics in young versus old mice shows that the replication stress-dependent inflammation might be aging associated. Inflammation is a hallmark of human aging,⁵ making it tempting to speculate that DNA replication stress contributes to this hallmark.

Limitations of the study

One limitation of this study is that we examined origin firing only in the context of the regenerating liver. We were limited to this system, because the EdU-seq-HU origin mapping method requires somewhat synchronous entry of cells in S phase. Whether the findings will apply to other tissues remains unknown. Mapping DNA replication origins in other organs will need to utilize methods, such as OK-seq and SNS-seq, that work best with unsynchronized cells.

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Thanos D. Halazonetis (thanos.halazonetis@unige.ch).

Materials availability

This study did not generate new unique reagents.

Data and code availability

- The fastq EdU-seq-HU, EdU-seq, and EU-seq sequencing files have been deposited at the NCBI BioProject and European Nucleotide Archive databases under accession code ENA: PRJNA732896. The fastq RNA-seq sequencing files have been deposited at the NCBI GEO DataSet database under accession number GEO: GSE231430. All the above data are publicly available as of the date of publication.
- This paper does not report original code. The scripts used to process and plot the sequencing data have been published in the supplementary section of reference Macheret and Halazonetis.²⁷
- Any additional information required to reanalyze the data reported in this paper is available from the [lead contact](#) upon request.

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AUTHOR CONTRIBUTIONS

Manipulation of mice, including performing partial hepatectomies and harvesting and purifying the hepatocytes, N.D., A.A.A., A.K., and D.S.; isolation of EdU-labeled DNA and EU-labeled RNA, high-throughput sequencing, and flow cytometry analysis, G.G.R. and A.K.; bioinformatics analysis, V.S.D., G.G.R., N.D., E.R.-C., and T.D.H.; imaging mass cytometry, T.Y. and A.A.A.; study concept and design, T.D.H., D.C., and D.S.; manuscript and figure preparation, T.D.H., D.S., and G.G.R., with contributions from all authors.

DECLARATION OF INTERESTS

T.D.H. is a Founder and Shareholder of FoRx Therapeutics AG and a member of its Scientific Advisory Board.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Ki-67 (1:200)	ThermoFisher	Cat# MA5-14520; RRID: AB_10979488
γ H2AX (1:1000)	Cell Signaling	Cat# 2577S; RRID: AB_2118010
Anti-rabbit biotinylated IgG (1:200)	ReactoLab	Cat# BA1000; RRID: AB_2313606
HNF- α (1:20) in house conjugated to 146Nd	Invitrogen	Cat# MA1-199; RRID: AB_2633309
CD19 (1:50) metal tag 142Nd	Standard BioTools	Cat# 3142014D; RRID: AB_2924313
F4/80 (1:50) in house conjugated to 152Sm	Abcam	Cat# AB240946; RRID: AB_10859466
CD3e (1:50) in house conjugated to 154Sm	BioLegend	Cat# 362702; RRID: AB_2563714
MHC class II (1:50) metal tag 161Dy	Standard BioTools	Cat# 91H038161
Arginase-1 (1:50) metal tag 164Dy	Standard BioTools	Cat# 3164027D; RRID: AB_2891145
T-bet (1:50) in house conjugated to 156Gd	Abcam	Cat# ab307194; RRID: AB_2938873
CD11b (1:100) metal tag 149Sm	Standard BioTools	Cat# 3149028D; RRID: AB_2891189
CD4 (1:100) metal tag 165Ho	Standard BioTools	Cat# 91H031165
Ki-67 (1:100) metal tag 168Er	Standard BioTools	Cat# 3168022D; RRID: AB_2811061
TIMD4 (1:100) in house conjugated to 172Yb	Thermo Fisher	Cat# PA5-116045; RRID: AB_2900679
Pan-actin (1:100) metal tag 175Lu	Standard BioTools	Cat# 3175032D
ICSK1 (1:100) metal tag 195Pt	Standard BioTools	Cat# 201500
ICSK2 (1:100) metal tag 196Pt	Standard BioTools	Cat# 201500
ICSK3 (1:100) metal tag 198Pt	Standard BioTools	Cat# 201500
Ly6-G (1:150) in house conjugated to 159Tb	BioLegend	Cat# 127602; RRID: AB_1089180
Nestin (1:200) in house conjugated to 143Nd	Abcam	Cat# ab237036; RRID: AB_2909415
CXCR6 (1:200) in house conjugated to 145Nd	Thermo Fisher	Cat# PA5-33462; RRID: AB_2550842
CD44 (1:200) metal tag 153Eu	Standard BioTools	Cat# 3153029D; RRID: AB_2890141
E-cadherin (1:200) metal tag 158Gd	Standard BioTools	Cat# 3158029D; RRID: AB_2893074
FoxP3 (1:200) metal tag 162Dy	Standard BioTools	Cat# 91H032162
CD8a (1:300) metal tag 167Er	Invitrogen	Cat# 14-0808-82; RRID: AB_2572861
CD31 (1:300) in house conjugated to 169Tm	Abcam	Cat# ab225883; RRID: AB_2920881
Collagen type I (1:300) metal tag 173Yb	Standard BioTools	Cat# 91H018173
CD206 (1:400) in house conjugated to 160Gd	R&D	Cat# AF2535; RRID: AB_2063012
α -SMA (1:600) metal tag 141Pr	Standard BioTools	Cat# 3141017D; RRID: AB_2890139
CD3 (1:600) metal tag 170Er	Standard BioTools	Cat# 3170019D; RRID: AB_2811048
CD45 (1:800) metal tag 151Eu	Standard BioTools	Cat# 91H029151
Glutamine Synthetase (1:800) in house conjugated 155Gd	Sigma	Cat# G2781; RRID: AB_259853
γ H2AX (1:1000) in house conjugated to 147Sm	Cell Signaling	Cat# 60566; RRID: AB_2118009
CK-7 (1:1000) in house conjugated to 163Dy	Abcam	Cat# ab220804; RRID: AB_2783822
CD45R/B220 (1:1000) metal tag 176Yb	Standard BioTools	Cat# 91H036176
Cell-ID Intercalator -Ir- (1:200) metal tag 125Um	Standard BioTools	Cat# 201192B
Chemicals, peptides, and recombinant proteins		
Buprenorphine (Temgesic)	Indivior Schweiz AG	7680441000291
Isoflurane	Piramal Pharma	NDC 66794-017-25
Vicryl 4.0	Ethicon	V1224H
Saline solution	Braun	7680295547065
Fentanyl	Sintetica S.A	7680539870034
Midazolam (Dormicum)	Cito Pharma Services GmbH	7680588750042

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
Medetomidine (Medetor)	Virbac	Cat#304729
HBSS (Mg2+/Ca2+ free, 10 mM HEPES pH 7)	ThermoFisher	Cat#14170-088
EDTA	ThermoFisher	Cat#15575020
Collagenase IV	Sigma-Aldrich	Cat#C5138; CAS: 900-12-1
Collagenase 1A	Sigma-Aldrich	Cat#C9891; CAS: 9001-12-1
DMEM	ThermoFisher	Cat#31966-021
Advanced DMEM/F-12	ThermoFisher	Cat#12634-010
FBS	Cytiva	Cat#SV30160.03
Penicillin/streptomycin/glutamine	ThermoFisher	Cat#10378-016
Hygromycin B	Sigma-Aldrich	Cat#H3274
CaCl2	Sigma	Cat#C7902; CAS: 10035-04-08
DPBS	ThermoFisher	Cat#4190-094
Methanol	Merck millipore	Cat#106009; CAS: 67-56-1
TRIZOL	ThermoFisher	Cat#15596
EdU	Sigma-Aldrich	Cat#900584; CAS: 61135-33-9
HU	Sigma-Aldrich	Cat#H8627; CAS: 127-07-1
ATR inhibitor, BAY-1895344	LubioScience	Cat#S8666
ATM inhibitor, AZD0156	Selleckchem	Cat#S8375
Aphidicolin	Sigma-Aldrich	Cat#A0781
Nocodazole	Tocris	Cat#1228
Triton-X100	PanReac, Applichem	Cat#A1388,2500
Cleavable biotin-azide linker	Jena Biosciences	Cat#CLK-CSTM
Copper (II) sulfate (CuSO4)	Acros Organics	Cat#197711000; CAS: 7758-98-7
Sodium L-ascorbate	Sigma-Aldrich	Cat#A7631; CAS: 134-03-2
THPTA	Sigma-Aldrich	Cat#762342; CAS: 760952-88-3
Phenol/chloroform/isoamyl alcohol (25:24:1)	Sigma-Aldrich	Cat#P3803
Chloroform/isoamyl alcohol (24:1)	Sigma-Aldrich	Cat#CO549
Ethanol	Sigma-Aldrich	Cat#51976; CAS: 64-17-5
Dynabeads™ MyOne™ Streptavidin C1	ThermoFisher	Cat#65002
Tris base	Sigma-Aldrich	SKU10708976001; CAS: 77-86-1
Sodium chloride (NaCl)	Sigma-Aldrich	Cat#S3014; CAS: 7647-14-5
Tween20	PanReac, Applichem	Cat#A4974,0100
β-Mercaptoethanol	Sigma-Aldrich	Cat#M6250; CAS: 60-24-2
EU	Jena Biosciences	Cat#CLK-N002-10
Isopropanol	Sigma-Aldrich	Cat#563935; CAS: 67-63-0
Alexa Fluor™647 azide	ThermoFisher	Cat#A10277
Propidium iodide	Sigma-Aldrich	Cat#81845; CAS: 25535-16-4
RNase, DNase free	Roche	Cat#11119915001
Formaldehyde solution 4%	Grogg Chemie AG	Cat#G256
Weiger's iron hematoxylin solution	Sigma-Aldrich	Cat#HT1079-1SET
Xylol	Grogg Chemie AG	Cat#G990
Antigen retrieval citrate buffer	Sigma	Cat#C9999
Tris Buffered Saline	Sigma	Cat#T6066
Eukitt quick-hardening mounting medium	Sigma-Aldrich	Cat#03989-100ML
Critical commercial assays		
TruSeq ChIP Sample Prep Kit	Illumina	Cat#IP-202-1012
Click-iT Nascent RNA Capture Kit	ThermoFisher	Cat#C10365
TruSeq Stranded Total RNA with Ribo-Zero Gold	Illumina	Cat#RS-122-2301

(Continued on next page)

Continued

REAGENT or RESOURCE	SOURCE	IDENTIFIER
TruSeq Stranded mRNA	Illumina	Cat#RS-122-2101
Guava H2A.X Phosphorylation Assay Kit (including 5x Assay Buffer and Anti-Histone H2A.X-Alexa Fluor™488)	Luminex	Cat#CR204438
ReliaPrep™ RNA Cell Miniprep System kit	Promega	Cat#Z6111
Vectastain Elite ABC Kit	Vector laboratories	Cat#PK-7100

Deposited data

EdUseq	This study	ENA: PRJNA732896
EUseq	This study	ENA: PRJNA732896
RNAseq	This study	GEO: GSE231430

Experimental models: Cell Lines

Human: U2OS CycE TetOFF	Laboratory of J. Bartek	N/A
Human: RPE1	ATCC	#CRL-4000

Experimental models: Organisms/strains

C57BL/6Jrj Mice (3- months old)	Janvier Labs	SC-C57J-M
Aged C57BL/6Jrj Mice (12- months old)	Janvier Labs	SC-C57J-M
Aged C57BL/6Jrj Mice (18- months old)	Janvier Labs	SC-C57J-M
Aged C57BL/6Jrj Mice (22- months old)	Janvier Labs	SC-C57J-M

Software and algorithms

Prism 9.0	GraphPad	https://www.graphpad.com/scientific-software/prism/
Kaluza Analysis Software	Beckman Coulter	https://www.beckman.com/flow-cytometry/software/kaluza
Burrows–Wheeler aligner tool (v.0.7.17)	Open source software for analysis of DNA sequences	https://bio-bwa.sourceforge.net/
QuPath	Open source software for digital pathology image analysis	https://www.nature.com/articles/s41598-017-17204-5

Other

22G Catheter	BD Insyte	Cat# BD 381223
40 µm nylon filter	PluriSelect	Cat#SKU 43-57040-03
70 µm nylon filter	Falcon	Cat#352350
2100 BioAnalyzer	Agilent	N/A
Bioruptor Pico sonication device	Diagenode	Cat#B01060003
Gallios 8-clor/2-laser flow cytometer	Beckman Coulter	N/A
Pannoramic MIDI Scanner	3DHISTECH	N/A

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Animals

C57BL/6 mice were purchased from Janvier Labs. Mice age, weight and sex are reported in Table S1. The cohort of young mice, referred to as 3-month-old mice, comprised mice that were 12- to 18-weeks-old and which had an average weight of 21 g. The cohort of aged mice comprised mice that were 12-, 18- and 22-months-old, with an average weight of 32 g. All mice were housed under controlled conditions, at a temperature of 25 °C, in a 12 h light/12 h dark cycle and with free access to food and water. All experiments were approved by the Institutional Animal Care and Use Committee of the University of Bern, Switzerland, and were in strict accordance with good animal practice as defined by the Office of Laboratory Animal Welfare.

Human cell lines

RPE1 cells (ATCC) were cultured in Dulbecco's Modified Eagle's Medium/Ham's F-12 (Invitrogen), supplemented with 10% FBS, penicillin/streptomycin/glutamine (ThermoFisher) and hygromycin B (Sigma).

METHOD DETAILS

Partial hepatectomy

Young and old mice underwent a standard PH resecting two thirds of the liver. Briefly, 0.05 mg/kg buprenorphine (Indivior Schweiz AG) was intraperitoneally injected for analgesia prior to surgery. The mice were then anesthetized by isoflurane (Piramal) inhalation during surgery, immobilized in a supine position and the liver exposed by transverse laparotomy. Following a laparotomy, Vicryl 4.0 suture (Ethicon) was used to ligate the left and median liver lobes. The ligated liver lobes were then resected using surgical scissors. The peritoneal cavity was irrigated with saline solution (Braun) and the abdomen was closed using a two-layer running suture. Surgery lasted 20 to 25 minutes. The resected tissue was collected and used for histological assessment of the liver. After PH, the liver was allowed to regenerate for the indicated times, before proceeding with the isolation of hepatocytes or tissue collection. Except where specifically indicated, all experiments followed the same regimen for ATR or ATM administration: the mice received 25 mg/kg ATR inhibitor (LubioScience, BAY-1895344) or 25 mg/kg ATM inhibitor (Selleckchem, AZD0156) by oral gavage 24 h post-PH.

Hepatocyte isolation

Hepatocytes were isolated by *in vivo* perfusion of the liver with a two-step collagenase digestion. Mice were anesthetized by intraperitoneal injection of a triple mix of 50 μ g/kg fentanyl (Sintetica S.A), 5 mg/kg midazolam (Cito Pharma Services GmbH) and 500 μ g/kg medetomidine (Virbac Medetor) 20 minutes prior to perfusion. The mice were immobilized in a supine position and the liver exposed by midline laparotomy. The portal vein was cannulated with a 22G catheter (BD Insyte) and perfused at 2.5 ml/min with the buffers allowed to run to waste through an incision in the inferior vena cava. The liver was perfused with 10 ml of HBSS (Mg²⁺/Ca²⁺-free, 10 mM HEPES pH 7) and then with 25 ml of HBSS containing EDTA (10 mM HEPES pH 7, 5 mM EDTA). The EDTA was removed by an additional perfusion with 10 ml HBSS followed by digestion with 25 ml of HBSS containing collagenase IV (10 mM HEPES pH7, 1 mM CaCl₂, 0.5 mg/ml collagenase IV (Sigma-Aldrich), 0.01 mg/ml collagenase 1A (Sigma-Aldrich)). The liver was excised, and the cells released by cutting the capsule and gently agitating the digested liver in ice-cold stop buffer (DMEM with 10% FBS, 1 mM CaCl₂). The suspension of hepatocytes was passed through a 70 μ m filter and centrifuged at 50 x g for 5 min, 4 °C. The cell pellet was resuspended in 50 ml of ice cold DPBS (ThermoFisher) and centrifuged at 50 x g for 5 min, 4 °C. Lastly, the hepatocytes were either fixed with ice-cold 90% methanol for EdUseq, EdUseq-HU and flow cytometry analysis or lysed with TRIzol (ThermoFisher) for EUseq.

In vivo EdUseq and EdUseq-HU

To monitor nascent DNA synthesis, at various time points after PH, the mice were injected with 100 mg/kg EdU (Sigma-Aldrich) (EdUseq) or with 100 mg/kg EdU and 500 mg/kg HU (Sigma-Aldrich) (EdUseq-HU) and hepatocytes were isolated 2 h later (Tables S1A and S1B). ATR or ATM inhibitors were also administered by gavage 24 h post-PH, as indicated. Sham-operated mice, injected with EdU and HU, served as no-hepatectomy controls. Hepatocytes were isolated and fixed, as described above. After fixation, the hepatocytes were permeabilized with 0.2% triton-X in PBS. EdU was coupled to a cleavable biotin-azide linker (Jena Biosciences) by a click-it reaction (100 mM sodium phosphate pH 7; 2 mM CuSO₄; 500 μ M Biotin; 10 mM sodium-L-ascorbate; 12 mM THPTA). Then, the DNA was purified by standard phenol-chloroform extraction followed by ethanol precipitation and sonicated to a size range of 100–500 bp with a bioruptor sonicator (Diagenode). The EdU-labelled DNA fragments were then isolated, as previously described.^{27,34} In brief, beads were washed three times in 1X Binding and Washing Buffer (5 mM Tris-HCl, pH 7.5, 0.5 mM EDTA, 1 M NaCl, 0.5% Tween 20). After the washes, the beads were resuspended in 1X Binding and Washing Buffer containing the sonicated DNA (5–10 μ g) and incubated for 15 min on a rotating wheel at room temperature. Beads were washed three times with 1X Binding and Washing Buffer and once with TE (10 mM Tris-HCl pH 8, 1 mM EDTA). EdU-labelled DNA fragments were eluted with 2% β -mercaptoethanol (Sigma-Aldrich) in TE for 1 h at room temperature and then used for library preparation, which was performed by the Genomics Platform of the University of Geneva using the TruSeq ChIP Sample Prep Kit (Illumina). Single-end read sequencing reactions (100 bp) were performed on Illumina Hi-Seq 2500 or Illumina Hi-Seq 4000 sequencers.

EdUseq-HU in cell lines

To induce mitotic arrest, RPE1 cells were treated with 200 ng/ml nocodazole (Tocris) for 8 h; the mitotic cells were isolated by shake-off, washed with PBS, released in warm medium containing 2 mM HU and 25 μ M EdU, and then collected 14 h later and fixed with 90% methanol overnight (Table S1C). After fixation, EdU was coupled to a cleavable biotin-azide linker; the samples were then processed as described above.

In vivo EU-seq

To monitor nascent transcription at various times after partial hepatectomy, the mice were injected with 134 mg/kg EU (Jena Biosciences) 30 min before hepatocyte isolation (Table S1D). RNA was extracted and purified with TRIzol and isopropanol precipitation and nascent EU-labelled RNA was biotinylated, purified and eluted as previously described.^{27,34} Briefly, nascent RNA was biotinylated and purified using the reagents of the Click-iT Nascent RNA Capture Kit (ThermoFisher). Dynabeads MyOne streptavidin C1 (ThermoFisher) were used to capture the EU-labelled RNA; the beads were washed three times with 1X Binding and Washing Buffer (5 mM Tris-HCl pH 7.5, 0.5 mM EDTA, 1 M NaCl, 0.5% Tween-20), two times with 0.1 M NaOH, 0.05 M NaCl for 2 min and two times

with 0.1 M NaCl for 2 min. The RNA, after being heated at 70 °C and placed back on ice, was incubated with the beads in 2X Binding and Washing Buffer for 30 min on a rotating wheel at room temperature. The beads were then washed three times with 1X Binding and Washing Buffer and once with RNase-free water. EU-labelled RNA was eluted with 2% β -mercaptoethanol (Sigma-Aldrich) for 1 h at room temperature. The EU-labelled RNA was used for library preparation, which was performed by the Genomics Platform of the University of Geneva using TruSeq Stranded Total RNA with Ribo-Zero Gold (Illumina) without the ribo-depletion step. Single-end read sequencing reactions (100 bp) were performed on Illumina Hi-Seq 2500 or Illumina Hi-Seq 4000 sequencers.

RNAseq

RNAseq was performed with RNA isolated from mouse liver lobes harvested during the PH or at 48 and 72 h post-PH (Table S1E). Where indicated, ATR inhibitor, at a dose of 25 mg/kg, was administered by oral gavage 24 h post-PH, except for the mice that were sacrificed 48 h post-PH, which received an additional 12.5 mg/kg dose of ATR inhibitor 36 h post-PH. The livers were harvested by excision. RNA was isolated using the ReliaPrep™ RNA Cell Miniprep System kit (Promega) following the manufacturer's instructions from snap-frozen liver tissue. RNA quality and concentration were measured with a 2100 BioAnalyzer (Agilent). The isolated RNA was used for library preparation, performed by the Next Generation Sequencing (NGS) Platform of the University of Bern using TruSeq Stranded mRNA (Illumina). Paired-end read sequencing reactions (50 bp) were performed on an Illumina NovaSeq6000 sequencer.

Immunohistochemistry

Immunohistochemistry was performed on tissue resected at the time of PH or on regenerating liver tissue from mice sacrificed 48 h post-PH; these were the same mice from which the 48 h post-PH RNA samples were isolated for RNAseq analysis (Tables S1E and S1F). The harvested liver lobes were fixed for 24 h in 4% formaldehyde solution (Grogg Chemie) and then processed for histological assessment. Paraffin-embedded tissues were sectioned at a thickness of 3.5 μ m and incubated with primary antibody against Ki67 (Thermo Fisher) or γ H2AX (Cell Signaling), followed by a secondary anti-rabbit IgG biotinylated antibody (Vectastain Elite ABC Kit (Vector laboratories) was used for signal amplification. Tissues were counterstained with Weiger's iron hematoxylin solution (Sigma) and slides were mounted with Eukitt (Sigma). All slides were scanned using the 3DHISTECH Panoramic MIDI Scanner. For the quantification of the fraction of actively proliferating hepatocytes, the cells were size-sorted and Ki67-positive and negative hepatocytes were quantified manually using the program QuPath.

Imaging mass cytometry

Imaging mass cytometry (IMC) analysis was performed on histology sections from the mice used for the RNAseq experiments (Table S1E). Tissue sections, 3.5 μ m thick, were stained with a validated 32-target antibody panel, according to previously described methods.¹⁰³ Cell-ID Intercalator -Irr- (Standard BioTools) was used to stain cell nuclei. The tissue slide was processed on the Hyperion™ mass cytometer (Standard BioTools). Imaging data were visualized using napari¹⁰⁴ and processed as described in the Steinbock pipeline, except for cell segmentation masks being produced using Cellpose TissueNet model.^{105,106} Processed data were analyzed using R v.4.3.06 in RStudio following the "IMC Data Analysis Workflow" as described at <https://bodenmillergroup.github.io/IMCDataAnalysis>, skipping the spillover and batch effect correction steps. Briefly, using markers associated with specific cell types, hepatocytes, liver sinusoidal endothelial cells (LSEC), biliary epithelial cells (BEC), vessel cells, Kupffer cells (KC), other generic macrophages, neutrophils, CD4-positive cells, CD8-positive cells and B cells were annotated using an in-house app. Hepatocytes were further separated into 4 clusters, which allowed us to distinguish periportal hepatocytes (clusters 1, 2) from pericentral hepatocytes (clusters 3, 4) using Glul intensity levels. Based on Ki67 intensity levels, clusters 2 and 4 were assigned as proliferating. Generic bone-marrow-derived macrophages were similarly separated into 5 groups using channels CD11b, F4/80, MHCII, Ki67, CD206, and CD45 for clustering. Clusters 1, 2 and 5 were observed to have both CD11b and CD206 (assigned Macrophages), while clusters 3 and 4 were observed to contain CD11b without CD206 (assigned Monocytes). Based on Ki67 intensity levels, clusters 2 and 3 were assigned as proliferating. Labeled cells were then used to perform differential abundance testing using edgeR (R package v. 3.42.2), as described in chapter 6 of "Multi-Sample Single-Cell Analyses with Bioconductor" at <https://bioconductor.org/books/3.13/OSCA.multisample/>. An abundance matrix of the cell counts per sample (columns) per cell type (rows) was used as an input for the analysis.

Flow cytometry assessment of γ H2AX content and EdU incorporation

Hepatocytes were isolated at various times post-PH from mice that had been injected with 100 mg/kg EdU (Sigma-Aldrich) 2 h before being sacrificed. Some mice also received 25 mg/kg ATR inhibitor (BAY-1895344; LubioScience), which was delivered by oral gavage 24 h post-PH. Purified hepatocytes were obtained as described for the EdUseq protocol and then passed through a 40 μ m nylon filter (PluriSelect) to obtain a single cell suspension.

U2OS human osteosarcoma cells, expressing Cyclin E in an inducible manner,²⁷ were cultured without doxycycline for two days to induce Cyclin E overexpression; the cells were then exposed to 100 ng/ml nocodazole (Tocris) for 10 h to induce mitotic arrest. Cells in mitosis were isolated by shake-off, washed with PBS, released in warm medium containing 25 μ M EdU (Sigma-Aldrich) and, optionally, 0.4 μ M aphidicolin (Sigma-Aldrich) and/or 400 nM ATR inhibitor (BAY-1895344; LubioScience). The cells were then collected at the indicated time points and fixed with 90% methanol overnight.

A single cell suspension of either hepatocytes or U2OS cells was processed for flow cytometry using the Guava H2A.X Phosphorylation Assay Kit (Luminex) according to the manufacturer's instructions with minor modifications. In brief, the cells were centrifuged at 370 x g for 5 min and permeabilized with Permeabilization Buffer (Luminex) for 20 min on ice. After one wash step in 1x Assay Buffer (Luminex), the cells were resuspended in 96 μ l of 1x Assay Buffer and 4 μ l of Anti-Histone H2A.X-Alexa Fluor™488 (Luminex) and incubated for 1 h on ice in the dark. The cells were then washed with 1x Assay Buffer, resuspended in 200 μ l of click reaction cocktail (100 mM Tris pH8; 4 mM CuSO₄; 100 mM sodium-L-ascorbate; 1.4 μ g/ml Alexa Fluor™647 azide (ThermoFisher)) to label EdU and incubated for 30 min in the dark at room temperature. After a last wash in 1x Assay Buffer, the genomic DNA was stained with 5 μ g/ml propidium iodide (SigmaAldrich) in combination with 1 μ g/ml RNase (Roche). The flow cytometry profiles were acquired (30'000 to 50'000 events) with a Gallios flow cytometer (Beckman Coulter) monitoring γ H2AX (Alexa Fluor™488), EdU (Alexa Fluor™647) and DNA content (propidium iodide). Data of young and old mice were compared using an ANOVA test.

EdUseq, EUseq and RNAseq data processing

Sequencing reads were aligned on the mouse reference genome assembly NCBI Build GRCm38/mm10 using the mem algorithm of the Burrows–Wheeler aligner tool (v.0.7.17), retaining only the reads with the highest mapping quality score of 60. Data processing and visualization were performed using our previously published workflow of custom Perl scripts,²⁷ adjusted to process mouse data.

QUANTIFICATION AND STATISTICAL ANALYSIS

Sample sizes were not predetermined, and normal data distributions were assumed. For each experiment, the mice were randomly assigned into groups; the experiments with ATR inhibitors always included control-treated mice in the same experiment. The results of all the mice included in the study are reported. The statistic tests and P values are reported in the results section, figure legends or figures.

Supplemental figures

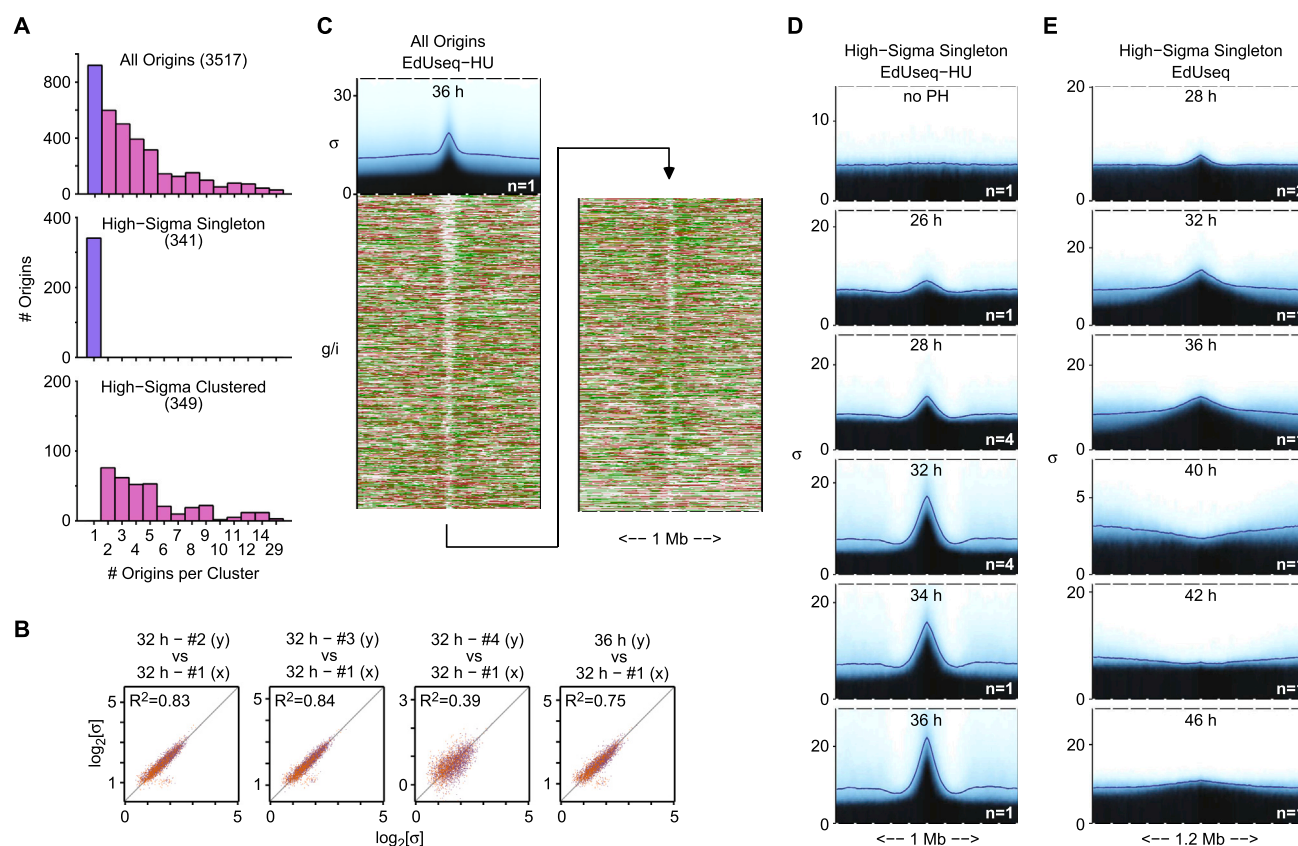


Figure S1. Profile of DNA replication origin firing in hepatocytes of regenerating livers of young adult mice, related to Figure 1

(A) Distribution of origins into groups according to clustering and peak height. Top: distribution of all origins according to whether they are singletons (blue) or part of clusters containing from 2 to 29 origins per cluster (purple). Middle: group of 341 singleton origins with peak height values greater than 15 sigma. Bottom: group of 349 clustered origins with peak height values greater than 25 sigma.

(B) Reproducibility of EdU-seq-HU data across different mice. Sigma (σ) values corresponding to all 3,517 origins are compared between different mice sacrificed 32 or 36 h post-PH, as indicated. Four mice were sacrificed 32 h post-PH and one 36 h post-PH. Coefficient of determination (R^2) values are shown for each comparison. Each dot corresponds to one origin; the dots for singleton and clustered origins are colored orange and purple, respectively. y, y axis; x, x axis.

(C) Overlay plot of 36-h post-PH EdU-seq-HU signals at all 3,517 origins. The line indicates the average sigma value. The heatmap below the EdU-seq-HU plot shows the annotated protein-coding genes, colored according to the direction of transcription: green, forward; red, reverse; white, intergenic regions. The genomic regions in the heatmap were plotted from top to bottom in order of decreasing EdU-seq-HU peak sigma value. Span of genomic regions shown: 1 Mb; bin resolution: 10 kb; ruler scale: 100 kb; n, number of mice from which the data were obtained.

(D) Overlay plots of EdU-seq-HU signals at the 341 high-sigma singleton origins in mice sacrificed at the indicated time points post-PH or after sham hepatectomy (no PH). The data are plotted as in (C).

(E) As (D), except that the mice were injected with EdU without HU. Span of genomic regions: 1.2 Mb.

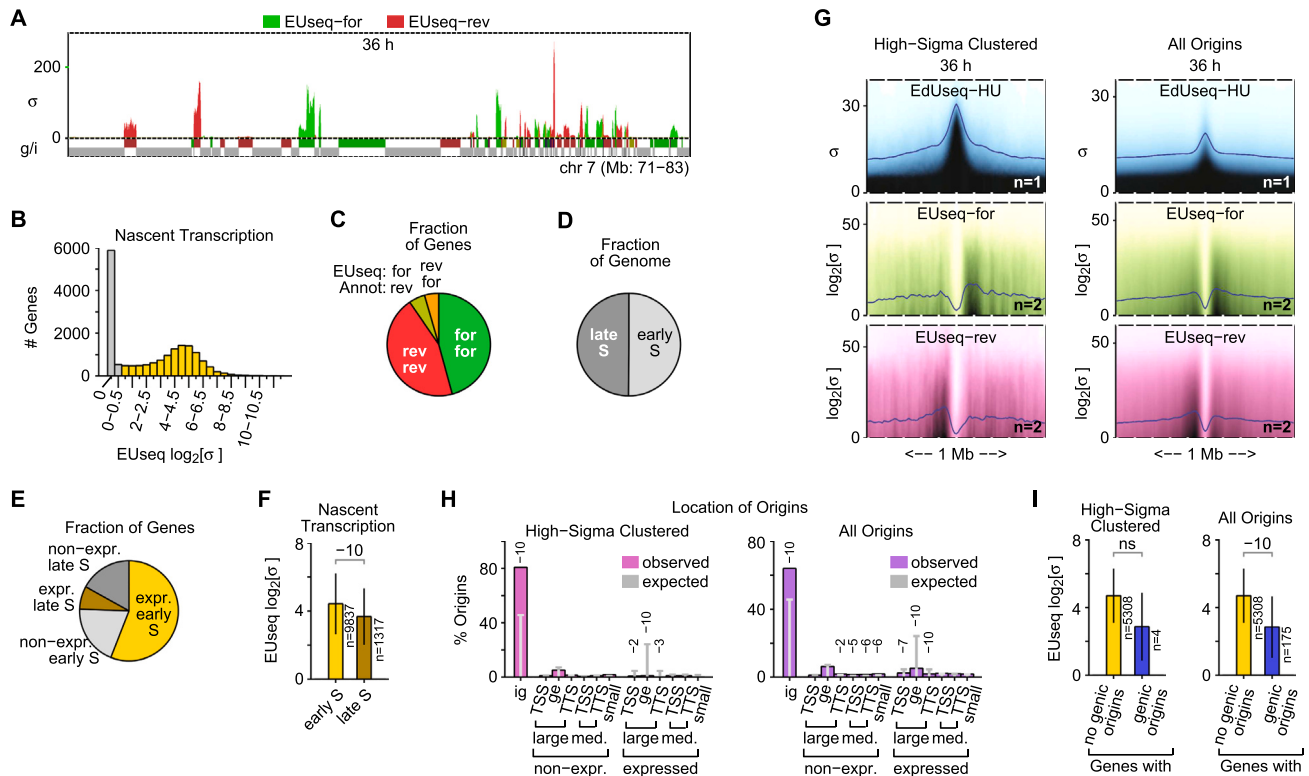


Figure S2. Profile of gene expression in relation to replication timing and location of replication origins in mouse regenerating livers, related to Figure 2

(A) Profile of *in vivo* incorporation of EU over the representative region of chr 7 (Mb: 71–83). The mice ($n = 2$) were sacrificed 36 h post-PH. The nascent transcripts are colored green and red to indicate forward (for) and reverse (rev) direction of transcription, respectively. σ , sigma value; g/i, gene/intergenic regions; bin resolution: 10 kb; ruler scale: 100 kb.

(B) Distribution of genes according to nascent transcription levels. For each gene, an average sigma value (σ) was calculated from the sigma values of all bins mapping to that gene. Genes with $\log_2[\sigma]$ values greater than 0.5 were considered as being expressed.

(C) Concordance of nascent transcription and gene annotation data. For each gene, the nascent EU-seq transcription data were used to determine the directionality of transcription as forward (for) or reverse (rev). The directionality of transcription was also obtained from gene annotation (Annot) data.

(D) Plot indicating the fraction of the mouse genome that is early or late replicating, as determined by the EdU-seq profiles 32 and 42 h post-PH (Figure 1C).

(E) The majority of expressed (expr.; defined as in B) genes map to early-replicating genomic domains.

(F) Comparison of nascent transcription levels of early-replicating versus late-replicating expressed genes. Means and standard deviations of EU-seq $\log_2$ sigma values are shown. The level of statistical significance is indicated as $\log_{10} p$ values; n , number of genes in each group.

(G) Overlay plots of 36-h post-PH EdU-seq-HU and EU-seq (forward and reverse directionality) signals at the 349 high-sigma clustered origins and at all 3,517 origins, plotted as in Figure 2B. The EdU-seq-HU incorporation data are from Figures 1F and S1C; n , number of mice from which the data were obtained.

(H) Distribution of the high-sigma clustered origins and of all origins on genomic bins, classified according to the scheme of Figure 2C. The expected frequencies refer to the fraction of each bin type in the early-replicating domains of the mouse genome. For bin types for which observed and expected frequencies were statistically different, the $\log_{10}$ of the p value is indicated. Bin classification codes: ig, intergenic; ge, genic; TSS, transcription start site; TTS, transcription termination site; small, small gene.

(I) Comparison of nascent transcription levels of large, early-replicating expressed genes with high-sigma clustered origins or any origin versus no origin mapping within their gene bodies. Means and standard deviations of EU-seq $\log_2$ sigma values are shown. The level of statistical significance is indicated by the $\log_{10}$ of the p value; ns, not significant; n , number of genes in each group.

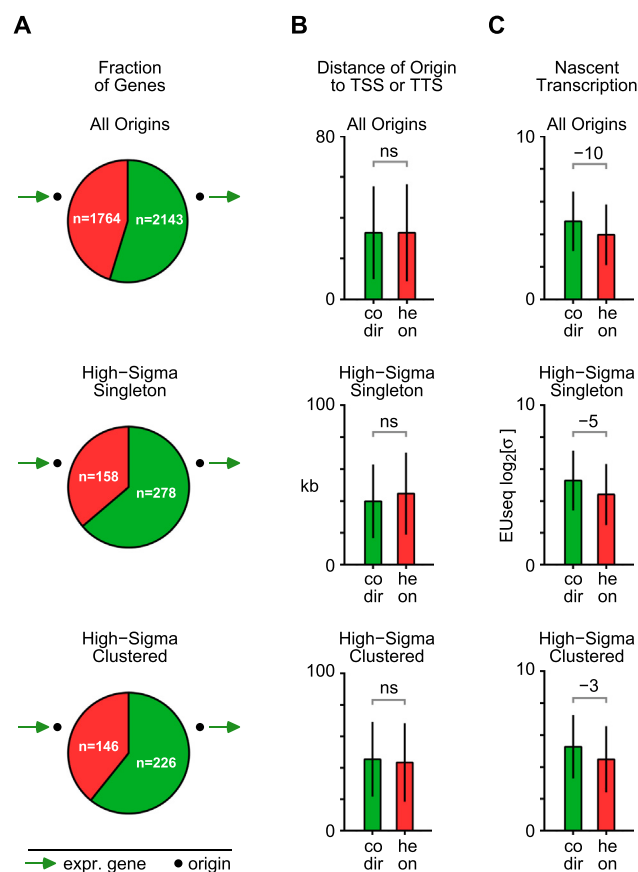


Figure S3. Directionality of replication versus transcription, related to Figure 3

(A) Fraction of genes replicated co-directionally or head-on with transcription from the list of early-replicating expressed genes that have only one replication origin in direct proximity (as defined in Figure 3A). Top, genes replicated by one origin of any type (this is the set of genes associated with one origin shown in Figure 3A); middle, genes replicated by one origin, which is a high-sigma singleton origin; bottom, genes replicated by one origin, which is a high-sigma clustered origin. *n*, number of genes in each group.

(B) Comparison of the distances of origins from the TSS or TTS for genes replicated co-directionally (codir) or head-on (heon) with transcription, respectively. Means and standard deviations of distances (kb) are shown. ns, not significant.

(C) Comparison of nascent transcription levels of genes replicated co-directionally or head-on with transcription. Means and standard deviations of EU-seq $\log_2$ sigma values are shown. The level of statistical significance is indicated as the $\log_{10}$ of the *p* values.

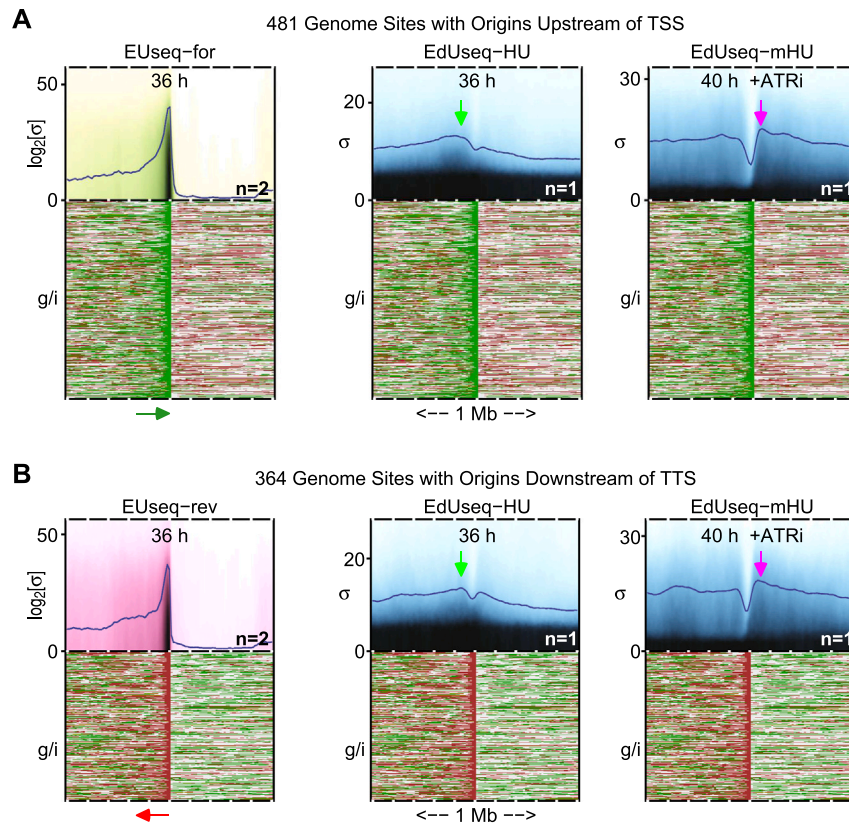


Figure S4. Firing of dormant origins, related to Figure 4

(A) Firing of dormant origins following treatment of mice with an ATR inhibitor 24 h post-PH and HU at 34, 36, and 38 h post-PH (multiple HU [mHU] administration) was monitored at 40 h post-PH and compared to a 36-h post-PH control with a single HU treatment at 34 h. Overlay plots of EU-seq and EdU-seq-HU signals at genomic regions containing genes with an origin of any type upstream of their TSS and no origin or expressed gene within 400 kb downstream of their TTS. The genes are plotted so that the direction of transcription is from left to right (horizontal, green arrow), and they are aligned by their TTS. The EU-seq and EdU-seq-HU datasets are from the same mice as in Figure 4B. The vertical green and purple arrows indicate the primary origins upstream of the TSSs and the dormant origins downstream of the TSSs, respectively. The data are plotted as in Figure 4B. *n*, number of mice from which the data were obtained.

(B) Firing of dormant origins illustrated using genomic regions containing genes with an origin of any type downstream of their TTS and no origin or expressed gene within 400 kb upstream of their TTS. The genes are plotted so that the direction of transcription is from right to left (horizontal, red arrow), and they are aligned by their TTS. The EU-seq and EdU-seq-HU datasets are from the same mice as in (A). The vertical green and purple arrows indicate the primary origins downstream of the TTSs and the dormant origins upstream of the TSSs, respectively. The data are plotted as in Figure 4B.

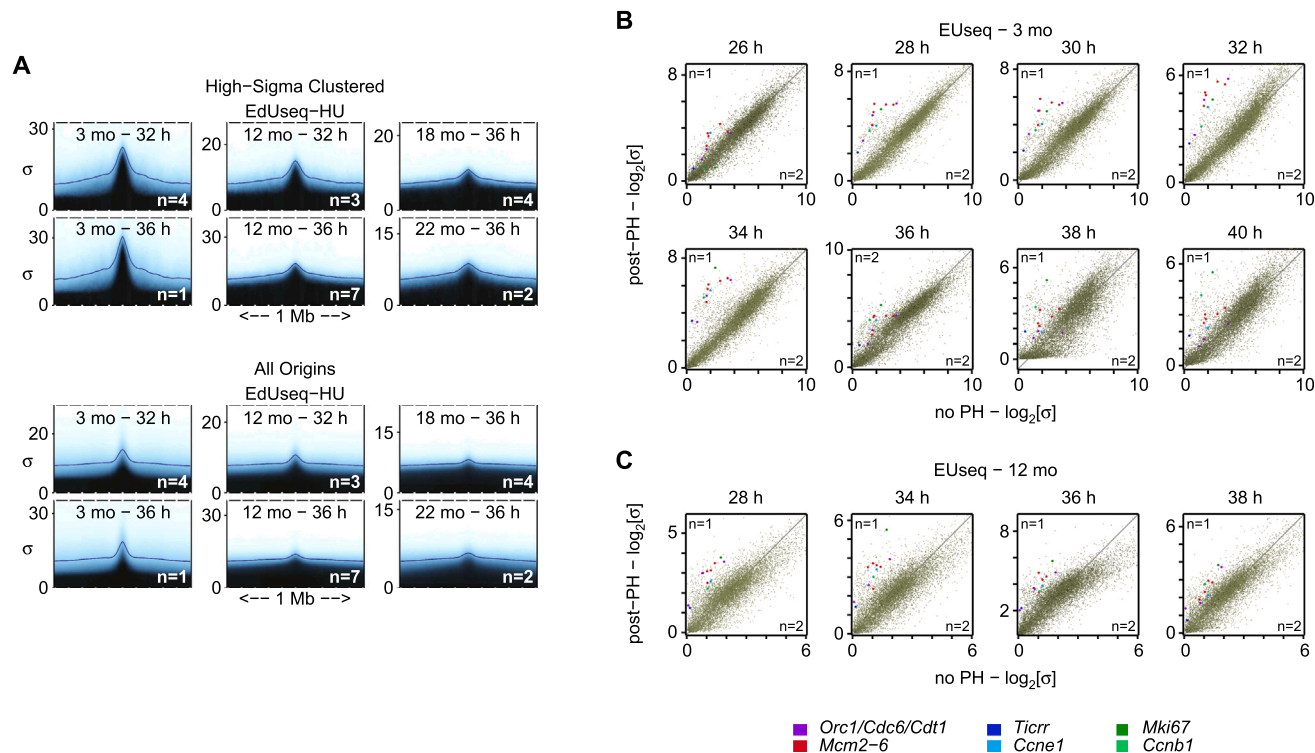


Figure S5. Defect in DNA replication origin firing with normal induction of gene expression in hepatocytes of aged mice after PH, related to Figure 5

(A) Overlay plots of EdU-seq-HU signals at the 349 high-sigma clustered origins or at all 3,517 origins in mice of the indicated ages 32 or 36 h post-PH. The data are plotted as in Figure 5B; *n*, number of mice from which the data were obtained. mo, month.

(B) Induction of gene expression in response to PH, monitored by EU-seq, before and at the indicated time points post-PH in 3-month-old mice. Each data point corresponds to one gene; the data points corresponding to select genes involved in DNA replication are color coded, as shown. σ , sigma value; *n*, number of mice from which the data were obtained.

(C) Induction of gene expression in response to PH, monitored by EU-seq, before and at the indicated time points post-PH in 12-month-old mice.

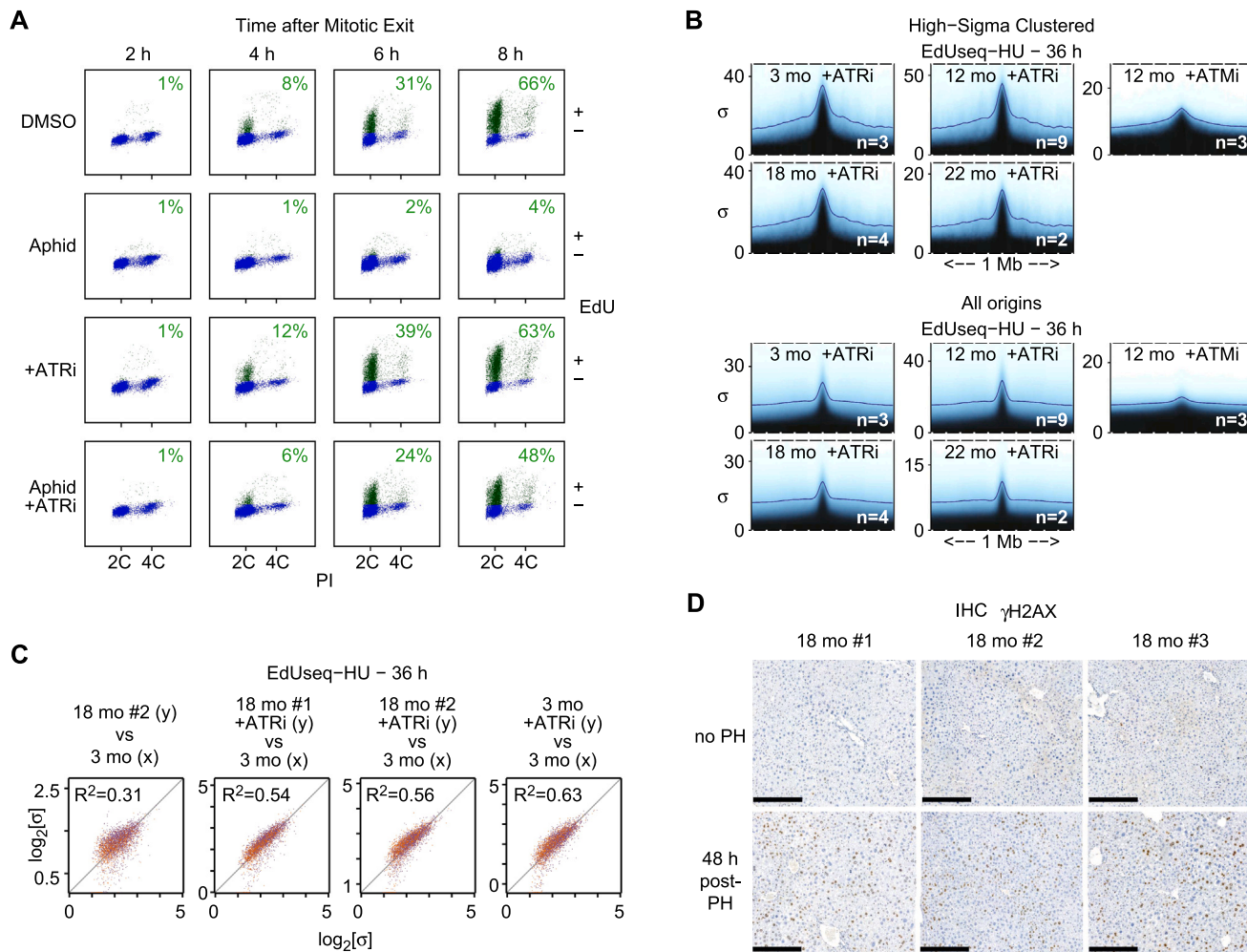


Figure S6. The DNA replication checkpoint suppresses origin firing in cells exposed to DNA replication stress and in aged mice, related to Figure 6

(A) Monitoring EdU incorporation of U2OS osteosarcoma cells exposed to aphidicolin (Aphid) and an ATR inhibitor (ATRi) during the G1 phase of the cell cycle and through entry into S phase. The cells were stained with propidium iodide (PI) for DNA content and analyzed by flow cytometry. The fraction of EdU-positive cells (green) is indicated.

(B) Enhanced origin firing efficiency in regenerating livers of aged mice treated with an ATR inhibitor (ATRi) but not an ATM inhibitor (ATMi). Overlay plots of EdU-seq-HU signals at the 349 high-sigma clustered origins or at all 3,517 origins 36 h post-PH in mice injected with ATRi or ATMi. The data are plotted as in Figure 6B; n , number of mice from which the data were obtained. mo, month.

(C) The enhanced origin firing efficiency induced by the ATR inhibitor in old mice affects all origins. EdU-seq-HU sigma (σ) values corresponding to all 3,517 origins from a young mouse are compared with the sigma values from an 18-month-old mouse, two 18-month-old mice treated with ATRi, and one young mouse treated with ATRi. All the mice were sacrificed 36 h post-PH. Coefficient of determination (R^2) values are shown for each comparison. Each dot represents one origin; dots for singleton and clustered origins are colored orange and purple, respectively. Lower $\log_2\sigma$ values on the y axis indicate a defect in origin firing. The data are from the same mice as shown in Figure 6A. y, y axis; x, x axis.

(D) Immunohistochemistry (IHC) staining for γ H2AX of liver tissue obtained at the time of PH (no PH) or 48 h post-PH from three 18-month-old mice. Scale bars: 200 μ m.

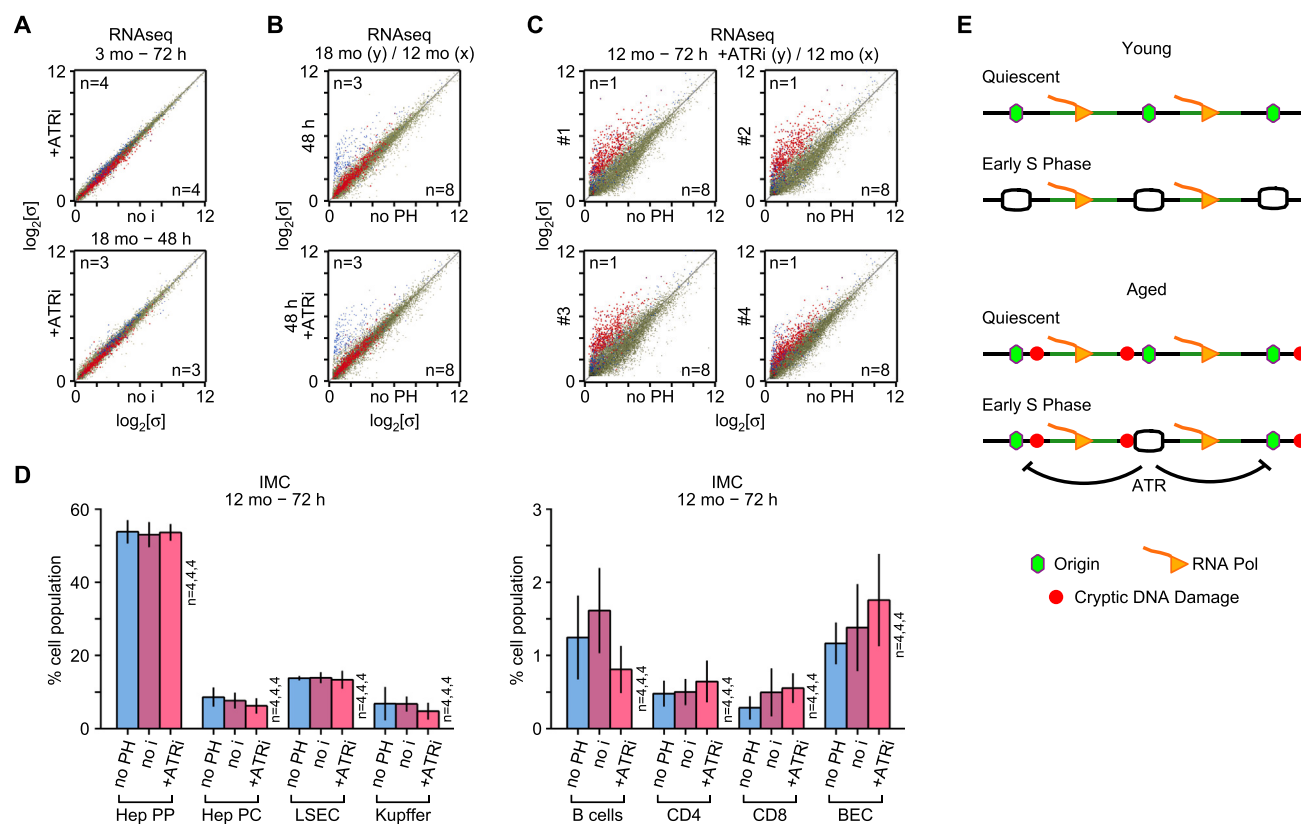


Figure S7. Induction of expression of cell-cycle and inflammatory response genes post-PH and cryptic DNA damage model of aging, related to Figure 7

(A) Treatment with ATR inhibitor (ATRi) does not affect gene expression in young mice 72 h post-PH or in old mice 48 h post-PH, as compared with mice not treated with the inhibitor (no i). Each data point corresponds to one gene and is colored by the selection criteria defined in Figure 7B. n , number of mice from which the data were obtained. mo, month.

(B) Induction of expression of cell-cycle genes in 18-month-old mice 48 h post-PH; the mice were untreated or treated with an ATR inhibitor. The no PH control mice were 12 months old. Each data point corresponds to one gene and is colored by the selection criteria defined in Figure 7B. n , number of mice from which the data were obtained; y, y axis; x, x axis.

(C) Induction of expression of inflammatory response genes 72 h post-PH by the ATRi in each of the four 12-month-old mice examined (#1, #2, #3, and #4). Each data point corresponds to one gene and is colored by the selection criteria defined in Figure 7B. n , number of mice from which the data were obtained; y, y axis; x, x axis.

(D) Fraction of periportal hepatocytes (Hep PP), pericentral hepatocytes (Hep PC), liver sinusoidal endothelial cells (LSECs), Kupffer cells, B cells, CD4 T cells, CD8 T cells, and biliary epithelial cells (BECs) in liver sections of aged mice, as determined by imaging mass cytometry (IMC). Liver tissues were obtained at the time the PH was performed (no PH) or at 72 h post-PH (no i, no inhibitor; ATRi, ATR inhibitor). Means, standard deviations, and number of mice in each group (n) are shown. PH and the administration of ATRi did not change the fraction of the above cell types, as compared with the no PH baseline.

(E) Cryptic DNA damage model of aging. In young mice, there is little accumulation of cryptic DNA damage. In old mice, cryptic DNA damage accumulates. In quiescent cells, this damage may have little effect on fitness. However, upon entry into S phase, the damage interferes with fork progression, leading to activation of a DNA replication stress response mediated by ATR, which then suppresses origin firing globally.